

MECHANICAL LEGEND

PIPING

— CWS —	CHILLED WATER SUPPLY
— CWR —	CHILLED WATER RETURN
— LCA —	COMPRESSED AIR PIPE
— HWS —	HEATING WATER SUPPLY
— HWR —	HEATING WATER RETURN
— HPS —	HIGH PRESSURE STEAM SUPPLY (700 kPa AND ABOVE)
— HPR —	HIGH PRESSURE STEAM RETURN
— LPS —	LOW PRESSURE STEAM SUPPLY (BELOW 200 kPa)
— LPR —	LOW PRESSURE STEAM RETURN
— CPD —	CONDENSATE PUMP DISCHARGE
— NG —	NATURAL GAS PIPE
— PW —	PLANT WATER

VALVES & SPECIALTIES

	FLOW INDICATOR
	SHUT-OFF VALVE
	BALL VALVE
	BUTTERFLY VALVE
	PRESSURE REDUCING VALVE
	GLOBE VALVE
	CHECK VALVE
	FLOW CONTROL VALVE W/ CHECK
	PLUG OR BALANCING SHUT-OFF VALVE
	N.O. VALVE WITH LOCKING COVER
	PLUG OR BALANCING SHUT OFF VALVE
	SHUT OFF VALVE IN RISER
	DRAIN VALVE W/ HOSE END
	STRAINER W/ BLOW-OFF VALVE
	TEMPERATURE CONTROL VALVE, 3-WAY
	TEMPERATURE CONTROL VALVE, 2-WAY
	PRESSURE REDUCING VALVE
	SAFETY OR RELIEF VALVE
	AIR VENT
	PRESSURE - TEMP. TAP
	PRESSURE GAUGE W/ PIG TAIL & COCK
	THERMOMETER
	VACUUM BREAKER
	PIPE EXPANSION JOINT
	FLEXIBLE PIPE CONNECTOR
	FLEXIBLE EQUIPMENT CONNECTOR
	PIPE UNION
	PIPE CAP
	PIPE ANCHOR
	PIPE GUIDE
	STEAM TRAP (SEE SCHEDULE FOR TYPE)
	PUMP
	ARROW INDICATES DOWNWARD PITCH OF PIPES
	PIPE ELBOW UP 90 OR 45
	PIPE ELBOW DOWN 90 OR 45
	PIPE TEE UP
	PIPE TEE DOWN

CONTROLS

	THERMOSTAT/SENSOR
	NIGHT THERMOSTAT
	HUMIDISTAT SENSOR
	STARTER
	PRESSURE SWITCH
	DIFFERENTIAL AIR PRESSURE SENSOR
	SWITCH
	FLOW METER
	LOW LIMIT THERMOSTAT
	DIFFERENTIAL PRESSURE GAUGE
	FLOW SWITCH

HVAC DUCTWORK

	FIRE DAMPER
	SMOKE DAMPER
	COMBINATION FIRE/SMOKE DAMPER
	TURNING VANE SHOWN IN 90° ELBOW
	SECTION THRU SUPPLY AIR DUCT
	SECTION THRU OA, RA OR EXH DUCT
	4 WAY SUPPLY DIFFUSER (UNLESS OTHERWISE NOTED)
	OPPOSED BLADE DAMPER
	BACKDRAFT DAMPER
	FLEXIBLE CONNECTION
	TEMPURATURE CONTROL DAMPER
	RETURN AIR GRILLE
	SUPPLY DIFFUSER

MISCELLANEOUS

	SECTION CUT
	POINT OF CONNECTION NEW TO EXISTING
	UNDERCUT DOOR 1"

NOTE: ALL SYMBOLS SHOWN ON LEGEND ARE NOT NECESSARILY USED

NIST

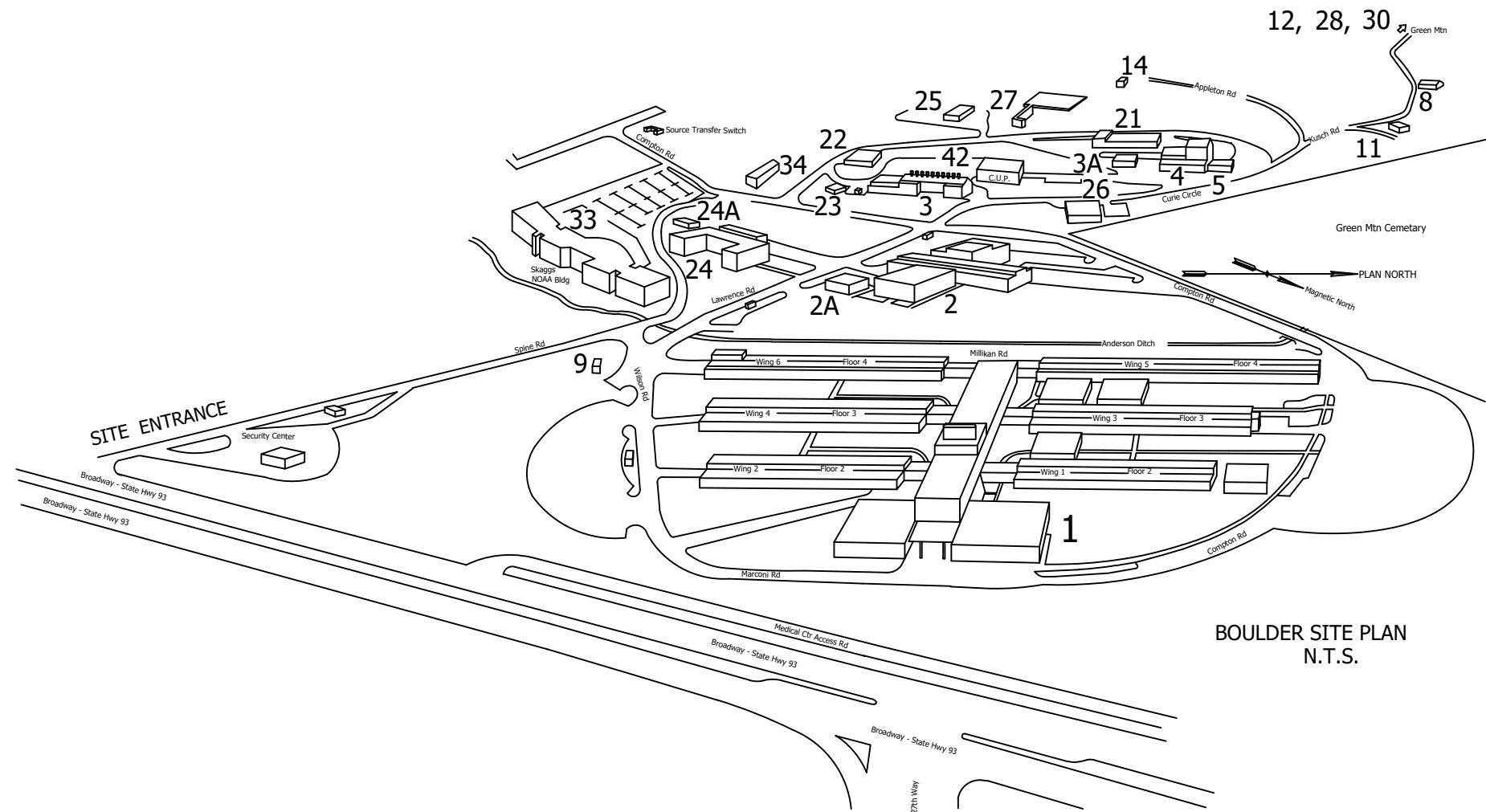
NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY

UNITED STATES DEPARTMENT OF COMMERCE

BOULDER COLORADO

ABBREVIATIONS

AFF	ABOVE FINISHED FLOOR	LF	LINEAR FEET
AFG	ABOVE FINISHED GRADE	LRA	LOCKED ROTOR AMPS
ALT	ALTITUDE	LWT	LEAVING WATER TEMPERATURE
BHP	BRAKE HORSEPOWER	MAX	MAXIMUM
BTU	BRITISH THERMAL UNIT	MCA	MINIMUM CIRCUIT AMPS
C	COMMON PORT	MBH	BTU PER HOUR (THOUSAND)
Cv	COEFFICIENT, VALVE FLOW	MIN	MINIMUM
CU FT	CUBIC FEET	NC	NOISE CRITERIA
CU IN	CUBIC INCH	N.O.	NORMALLY OPEN
CFM	CUBIC FEET PER MINUTE	N.C.	NORMALLY CLOSED
SCFM	CFM, STANDARD CONDITIONS	N/A	NOT APPLICABLE
dB	DECIBEL	NIC	NOT IN CONTRACT
DCW	DOMESTIC COLD WATER	NTS	NOT TO SCALE
DEG OR °	DEGREE	NO	NUMBER
DHW	DOMESTIC HOT WATER	OA	OUTSIDE AIR
DIA	DIAMETER	OD	OUTSIDE DIAMETER
DB	DRY-BULB	PPM	PARTS PER MILLION
DN	DOWN	%	PERCENT
EAT	ENTERING AIR TEMP.	PH OR Ø	PHASE (ELECTRICAL)
EFF	EFFICIENCY	PSF	POUNDS PER SQUARE FOOT
ELEV	ELEVATION	PSI	POUNDS PER SQUARE INCH
ESP	EXTERNAL STATIC PRESSURE	PSIA	PSI ABSOLUTE
(E)	EXISTING	PSIG	PSI GAUGE
EXIST	EXISTING	PRESS	PRESSURE
F	FAHRENHEIT	RA	RETURN AIR
FLA	FULL LOAD AMPS	RECIRC	RECIRCULATE
FPM	FEET PER MINUTE	RH	RELATIVE HUMIDITY
FPS	FEET PER SECOND	RLA	RUNNING LOAD AMPS
FT	FOOT OR FEET	RPM	REVOLUTIONS PER MINUTE
FU	FIXTURE UNITS	SL	SEA LEVEL
GA	GAUGE	SENS	SENSIBLE
GAL	GALLONS	SPEC	SPECIFICATION
GPH	GALLONS PER HOUR	SQ	SQUARE
GPM	GALLONS PER MINUTE	STD	STANDARD
HD	HEAD	SP	STATIC PRESSURE
HG	MERCURY	SA	SUPPLY AIR
HGT	HEIGHT	TEMP	TEMPERATURE
HORZ	HORIZONTAL	TD	TEMPERATURE DIFFERENCE
HP	HORSEPOWER	TSP	TOTAL STATIC PRESSURE
HR	HOURL(S)	TSTAT	THERMOSTAT
HWC	HOT WATER CIRCULATING (DOMESTIC)	TONS	TONS OF REFRIGERATION
HZ	HERTZ	VAV	VARIABLE AIR VOLUME
ID	INSIDE DIAMETER	VEL	VELOCITY
IE	INVERT ELEVATION	VERT	VERTICAL
IN	INCHES	V	VOLT
IN W.C.	INCHES WATER COLUMN	VOL	VOLUME
KW	KILOWATT	W	WATT
KWH	KILOWATT HOUR	WT	WEIGHT
LAT	LEAVING AIR TEMPERATURE	WB	WET-BULB



GENERAL NOTES:

- MANUFACTURER'S CATALOG NUMBERS ARE FOR SPECIFICATION PURPOSES. NIST ACCEPTED EQUIVALENTS SHALL BE ACCEPTED EXCEPT WHERE OTHERWISE INDICATED. SUPPLY NIST ENGINEERING WITH MFR'S CUT-SHEETS FOR APPROVAL PRIOR TO PURCHASE OF SUBSTITUTE EQUIPMENT.
- CONTRACTOR TO FIELD VERIFY EXISTING CONDITIONS PRIOR TO CONSTRUCTION.
- EXISTING MECHANICAL SYSTEMS SHOWN LIGHT LINE WEIGHT. NEW MECHANICAL SYSTEMS SHOWN HEAVY LINE WEIGHT. MECHANICAL SYSTEMS TO BE REMOVED ARE SHOWN CROSS-HATCHED OR DESCRIBED IN SYSTEM DEMOLITION NOTES.
- CONTRACTOR IS TO PROVIDE COMPLETE CONNECTIONS TO ALL NEW AND RELOCATED OWNER FURNISHED EQUIPMENT.
- CONTRACTOR SHALL MAKE ARRANGEMENTS FOR PARKING, COORDINATE ANY EQUIPMENT DOWNTIME, AND COORDINATE GENERAL WORK SCHEDULE WITH THE COR.
- THE DRAWINGS INDICATE THE GENERAL ARRANGEMENT OF THE EXISTING EQUIPMENT LOCATIONS, EXISTING PIPE ROUTING AND EXISTING DUCT ROUTING. CONTRACTOR IS TO INSTALL NEW EQUIPMENT, DUCTS, AND PIPING IN LOCATIONS REQUIRED TO AVOID INTERFERENCE WITH EXISTING FACILITIES, EQUIPMENT, DUCTS AND PIPING. CONTRACTOR IS RESPONSIBLE FOR RAISING AND LOWERING EXISTING EQUIPMENT AND MAKING PROPER OFFSETS IN DUCTS AND PIPING TO AVOID CONFLICTS.
- CONTRACTOR SHALL VISIT THE SITE BEFORE SUBMITTING BID, AND VERIFY DIMENSIONS AND EXISTING CONDITIONS RELATED TO THE WORK.
- MECHANICAL DRAWINGS ARE DIAGRAMMATIC AND DO NOT NECESSARILY INDICATE EVERY REQUIRED OFFSET, FITTING, ETC. DRAWINGS ARE NOT TO BE SCALED FOR DIMENSIONS. TAKE ALL DIMENSIONS FROM CERTIFIED EQUIPMENT DRAWINGS AND FROM THE STRUCTURE ITSELF BEFORE FABRICATING ANY WORK, VERIFY ALL SPACE REQUIREMENTS COORDINATING WITH OTHER TRADES, AND INSTALL THE SYSTEMS IN THE SPACE PROVIDED WITHOUT EXTRA CHARGES TO THE OWNER.
- ALL WORK SHALL BE INSTALLED IN ACCORDANCE WITH ALL APPLICABLE STATE CODES, LOCAL CODES AND OWNER'S STANDARDS INDICATED BY THE CONSTRUCTION DOCUMENTS.

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY
ENGINEERING, MAINTENANCE and SUPPORT SERVICES
325 BROADWAY BOULDER, COLORADO 80305-3328

BUILDING 2
Room 1330 Walk-In Fume Hood

MECHANICAL

DESIGN COOK	DRAFTING WIERSMA	APPROVAL	DATE	SHEET NUMBER M-0 of 1
PROJECT NAME B2 - Room 1330 Walk-In Fume Hood				

REVISION	DESCRIPTION	BY	SUPV.	DATE

MECHANICAL DEMO NOTES:

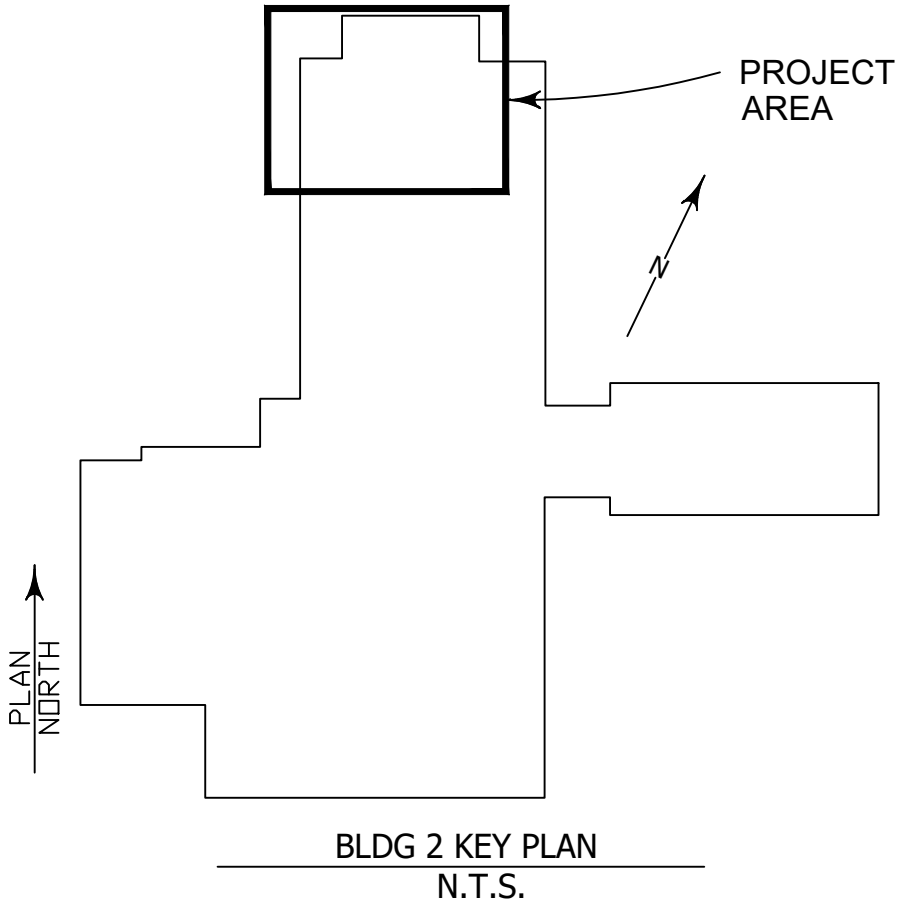
- 1

REMOVE FUME HOOD AND BASE CABINET COMPLETELY
- 2

REMOVE 10" PVC EXHAUST DUCT COMPLETELY, TEMPORARILY SEAL OPENING IN PLEXIGLAS.
- 3

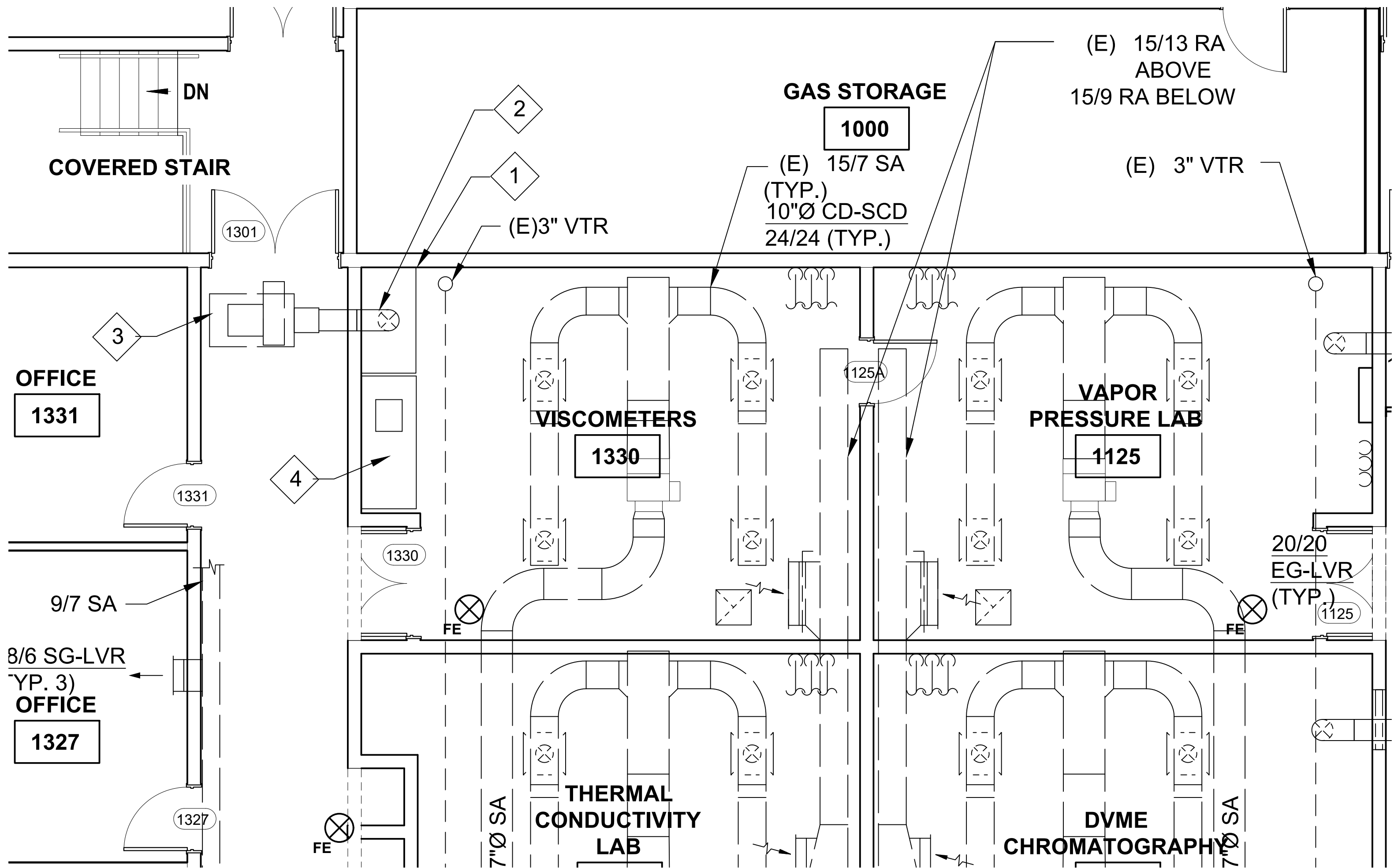
REMOVE EXHAUST FAN ON ROOF COMPLETELY, INCLUDING ALL SUPPORTS AND PVC DUCT .
- 4

REMOVE COUNTER TOP, BASE AND SINK COMPLETELY.DHW, DCW AND WASTE TO REMAIN FOR NEW SINK. ROUGH BLACK SINK BASIN UNDERCOATING IS 3% NON-FRIABLE ASBESTOS. CONTRACTOR SHALL DISPOSE OF THE SINK PER STATE AND FEDERAL REGULATIONS.

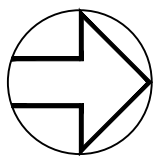


KEYNOTES

1. KEYNOTES



FIRST FLOOR DEMO PLAN
SCALE: 1/4" = 1'

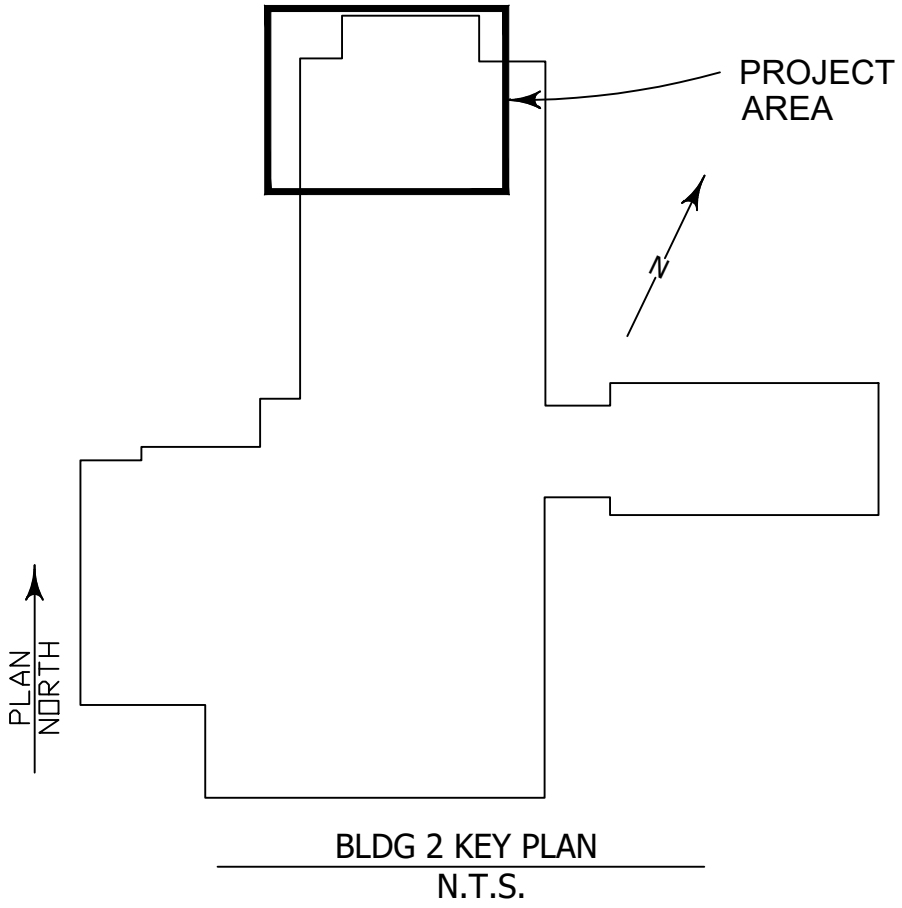


DESIGN	DRAFTING	APPROVAL	DATE	SHEET NUMBER
COOK	WIERSMA			M-1
PROJECT NAME				
B2 - Room 1330 Walk-In Fume Hood				
REVISION	DESCRIPTION	BY	SUPV.	DATE

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY ENGINEERING, MAINTENANCE and SUPPORT SERVICES 325 BROADWAY BOULDER, COLORADO 80305-3328				
BUILDING 2 Room 1330 Walk-In Fume Hood DEMOLITION MECHANICAL				
DESIGN	DRAFTING	APPROVAL	DATE	SHEET NUMBER
COOK	WIERSMA			M-1
PROJECT NAME				
B2 - Room 1330 Walk-In Fume Hood				
REVISION	DESCRIPTION	BY	SUPV.	DATE

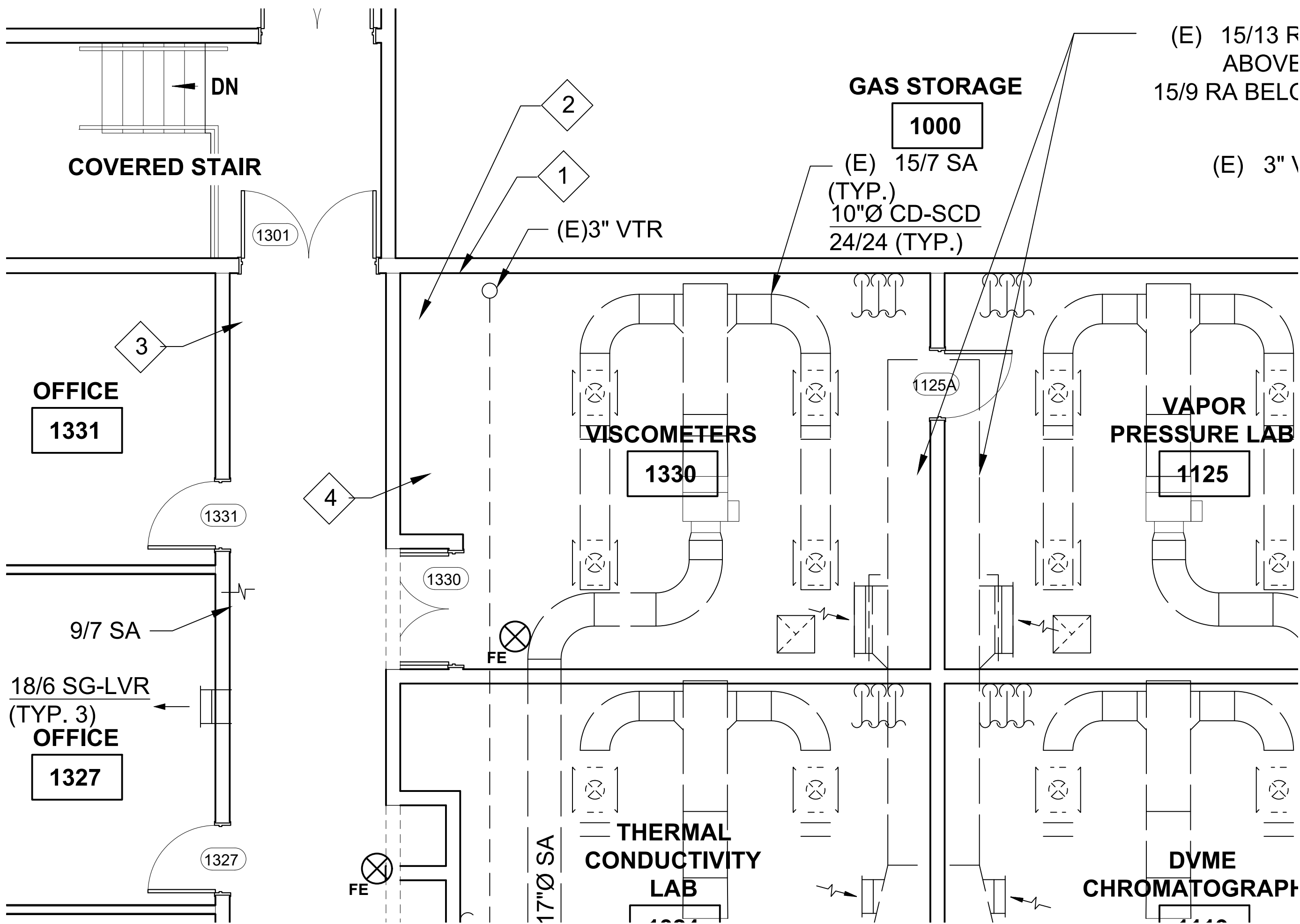
MECHANICAL NEW WORK NOTES:

- 1
- PROVIDE WALK-IN FUME HOOD 02-LFH-1330-1
- 2
- PROVIDE 12" PVC DUCTWORK, RE-USE & ENLARGE OPENING IN PLEXIGLAS. REPLACE PLEXIGAS IF EXISTING OPENING CAN NOT BE RE-USED.
- 3
- EXHAUST FAN 02-EF-LFH-1330-1 ON ROOF, REFER TO DWG M-4 & M-6.
- 4
- PROVIDE COUNTER TOP , SINK AND BASE PER SCHEDULE. CONNECT TO EXISTING DHW, DCW AND WASTE.

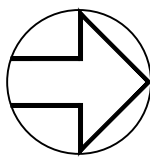


KEYNOTES

1. KEYNOTES

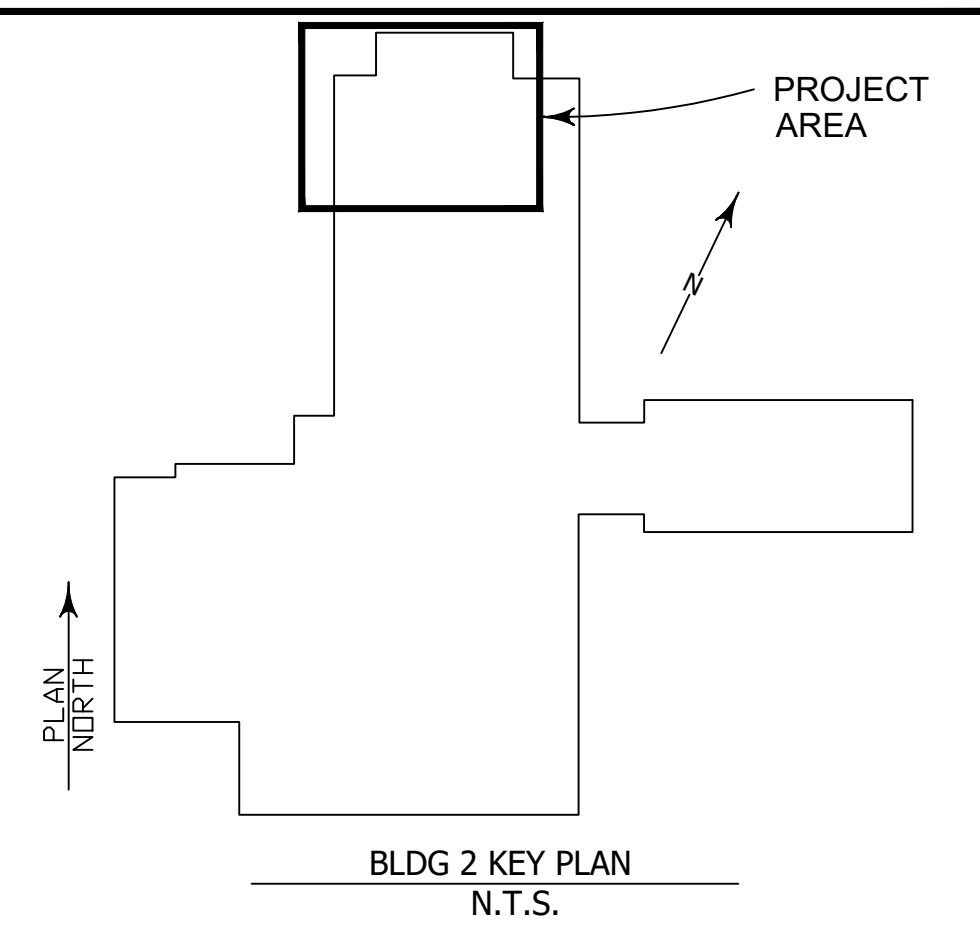


FIRST FLOOR PLAN
SCALE: 1/4" = 1'



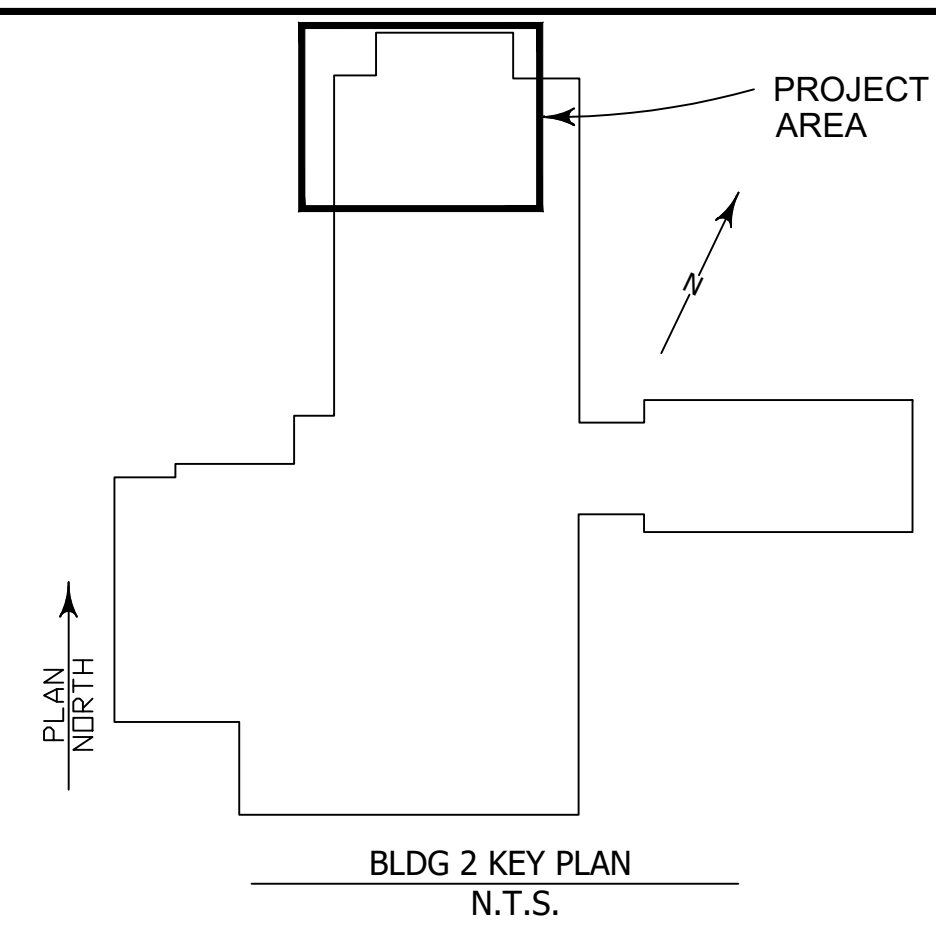
DESIGN	DRAFTING	APPROVAL	DATE	SHEET NUMBER
COOK	WIERSMA			M-2
PROJECT NAME				
B2 - Room 1330 Walk-In Fume Hood				

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY ENGINEERING, MAINTENANCE and SUPPORT SERVICES 325 BROADWAY BOULDER, COLORADO 80305-3328				
BUILDING 2 Room 1330 Walk-In Fume Hood NEW WORK MECHANICAL				
DESIGN	DRAFTING	APPROVAL	DATE	SHEET NUMBER
COOK	WIERSMA			M-2
PROJECT NAME				
B2 - Room 1330 Walk-In Fume Hood				



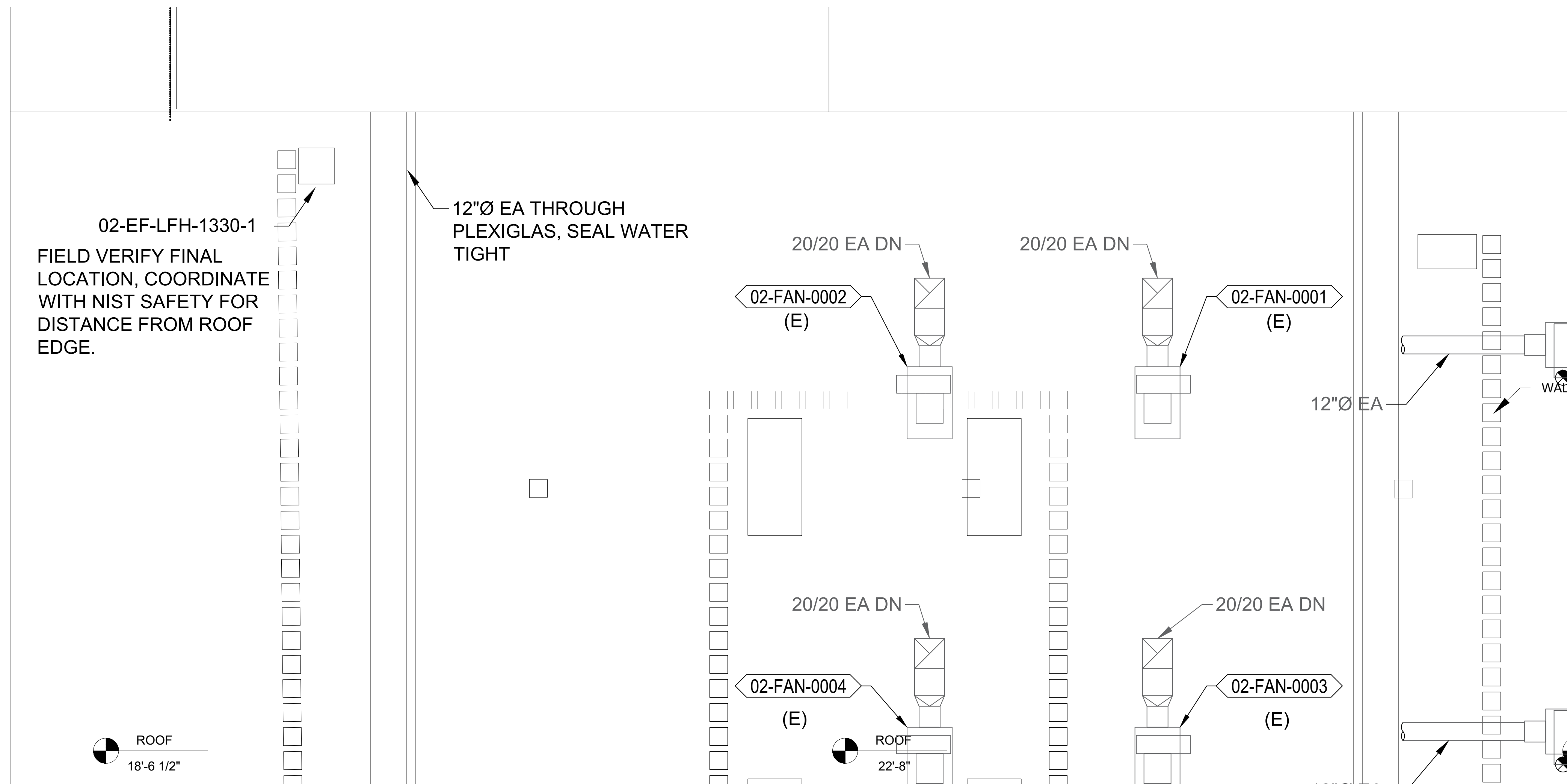
1. KEYNOTES

					MECHANICAL					
					DESIGN	DRAFTING	APPROVAL	DATE	SHEET NUMBER M-3 of	
					COOK	WIERSMA				
					PROJECT NAME					
REVISION	DESCRIPTION				BY	SUPV.	DATE	B2 - Room 1330 Walk-In Fume Hood		

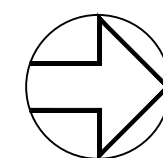


KEYNOTES

1. KEYNOTES



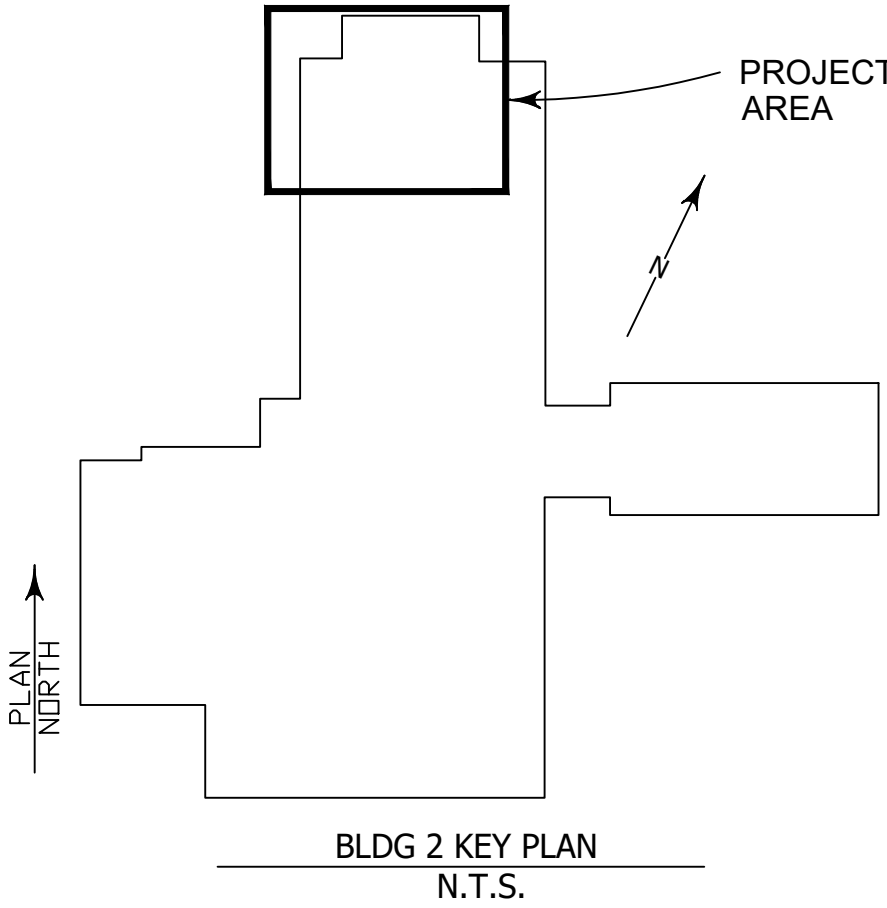
ROOF PLAN
SCALE: 1/4" = 1'



DESIGN	DRAFTING	APPROVAL	DATE
COOK	WIERSMA		
PROJECT NAME			
REVISION	DESCRIPTION	BY	SUPV. DATE

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY ENGINEERING, MAINTENANCE and SUPPORT SERVICES 325 BROADWAY BOULDER, COLORADO 80305-3328			
BUILDING 2 Room 1330 Walk-In Fume Hood ROOF PLAN MECHANICAL			
DESIGN	DRAFTING	APPROVAL	DATE
COOK	WIERSMA		
PROJECT NAME			
B2 - Room 1330 Walk-In Fume Hood			
SHEET NUMBER			M-4
			of

FAN SCHEDULE															
DESIGNATION	MANUFACTURER	MODEL	SERVICE	CFM AT 5430'	S.P. IN. W.C.	APPROX. RPM	NOZZLE OUTLET VELOCITY (FPM)	VOLTAGE/ PHASE	MIN. HP	MCA	MOP	VIBRATION ISOLATOR TYPE	DRIVE TYPE	WEIGHT (LBS)	SPEC NOTES
02-EF-LFH-1330-1	GREENHECK	FJ	HOOD 02-LFH-1330-1 EXHAUST	950	0.50	1,689	1,223	115/60/1	0.25	3.56	6	SPRING, RESTRAINED, 1 INCH	DIRECT	126	1-19
1.MOTOR VFD RATED WITHOUT SHAFT GROUNDING PROTECTION 2.CONTROL - DIAL FOR BALANCING 3.MOTOR - VARI-GREEN EC MOTOR, NEMA-3R MOTOR ENCLOSURE, FOR INDOOR/OUTDOOR USE, GALVANIZED 4.MOTOR WITH MINIMUM 40 DEGREE C AMBIENT TEMPERATURE 5.MOTOR WITH CLASS B OR GREATER INSULATION 6.FINISH - COATED 7.COATING - HI-PRO POLYESTER, CONCRETE GRAY-RAL 7023, FAN AND ATTACHED ACCESSORIES 8.SWITCH - NEMA-3R, TOGGLE, FOR INDOOR OR OUTDOOR USE, MOUNTED AND WIRED 9.UL LISTED FOR UL/CUL-705 - "POWER VENTILATORS" 10. DISCHARGE POSITION - UB 11. ARRANGEMENT - 4 12.ROTATION - CW 13.DRAIN CONNECTION - HOLE ONLY 14.ACCESS DOOR - BOLTED 15.INLET FLANGE, SLIP FIT 16.OUTLET CONNECTION - SLIP FIT 17.DIRECT MOUNT ISOLATORS, ISOLATOR-SPRING, RESTRAINED, 1 INCH, INDOOR / OUTDOOR USE 18.FASTENERS - STANDARD 19.UNIT WARRANTY: 1 YR															



KEYNOTES

1. KEYNOTES

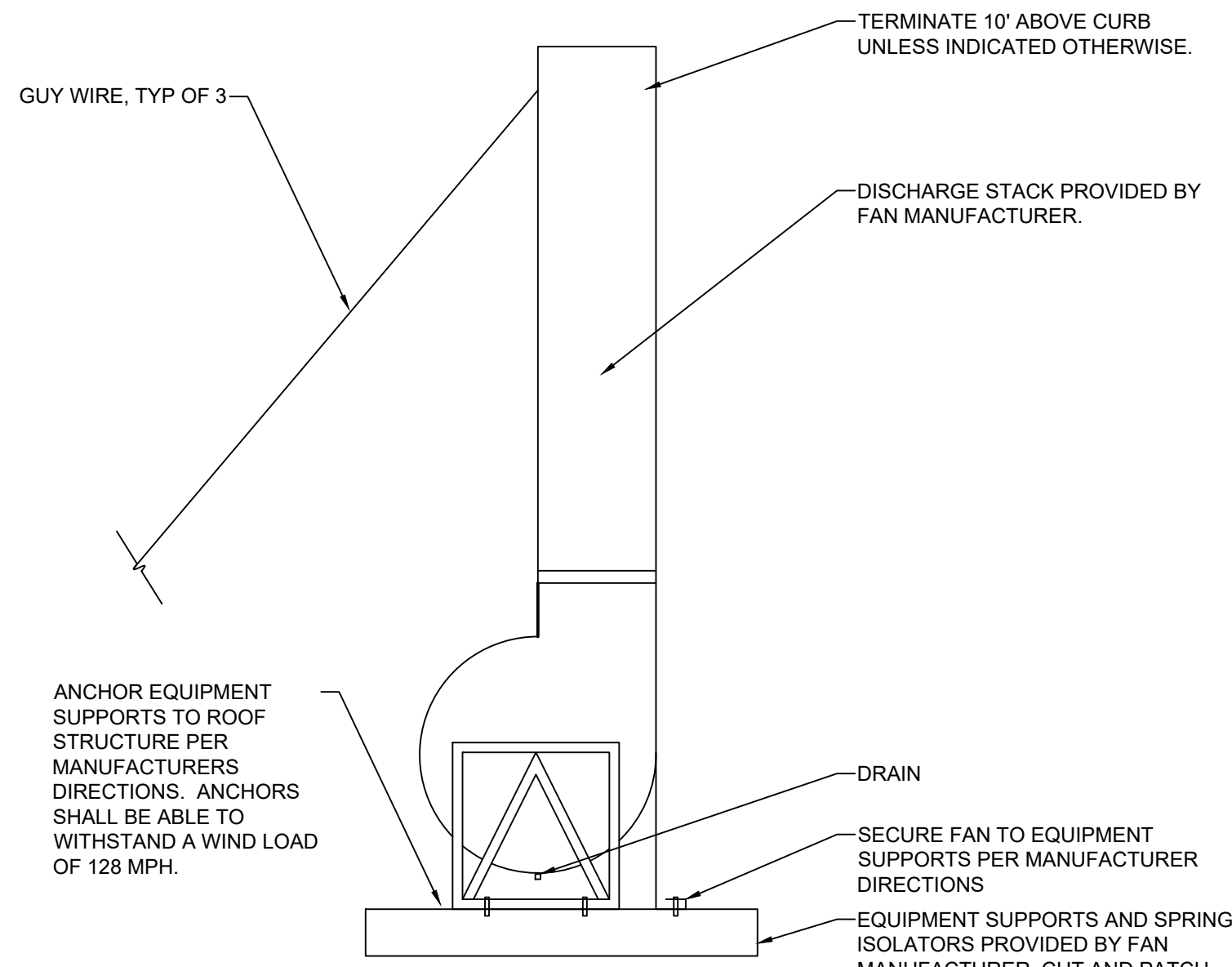
CASEWORK SCHEDULE				
DESIGNATION	MANUFACTURER	CASEWORK	FIXTURES	COLOR
CW-1	BEDCOLAB	(1) - 4' WIDE INSET STEEL SINK BASE CABINET. (1) - 3' WIDE MODESTY PANEL WITH TABLE RAILS AND END PANEL.	(1) - DECK MOUNTED H/CW MIXING FAUCET WITH VACUUM BREAKER AND BLADE HANDLES. (1) - DECK MOUNTED EYE WASH WITH DRENCH HOSE. GAURDIAN MODEL G50228P OR EQUAL (1) - THERMOSTATIC MIXING VALVE, GAURDIAN MODEL G6040 OR EQUAL	TO BE SELECTED BY COR

INSTANT DOMESTIC WATER HEATER SCHEDULE							
DESIGNATION	MANUFACTURER	MODEL	WATT INPUT	ELECTRICAL		WATER TEMP. RISE (°F)	NOTES
				VOLTS	PHASE		
02-WH-1330-01	CHRONOMITE	SR-20L/208	4,160.0	208	1	28	

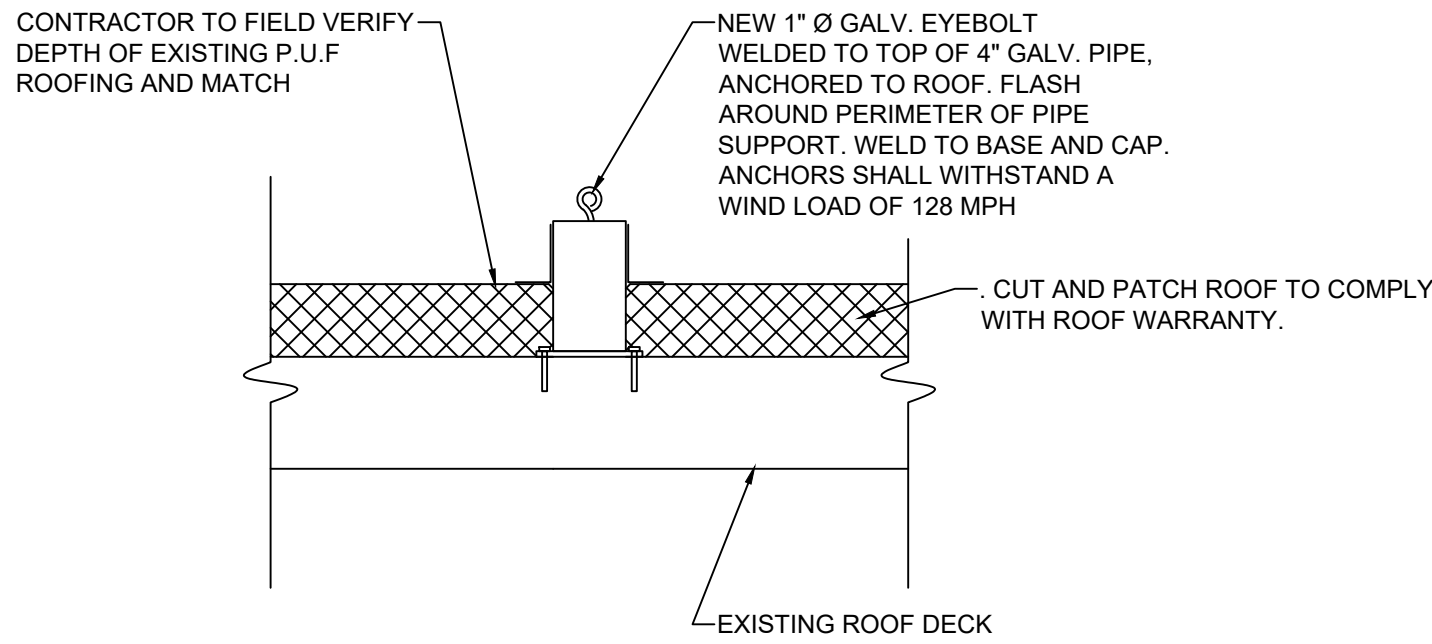
FLOOR MOUNTED FUME HOOD SCHEDULE												
DESIGNATION	MANUFACTURER	MODEL	SASH TYPE	INTERIOR DEPTH	EXHAUST WITH 27.25" FULL OPEN SASH	EXHAUST WITH 18" OPEN SASH	EXHAUST CONNECTION SIZE	FLOOR	LINER	LINER	COLOR	SPEC NOTES
02-LFH-1330-1	BEDCOLAB	6' VANGUARD FLOOR-MOUNTED HOOD (#C71-72C-400)	VERTICAL	40.75"	946 CFM @ 80 FPM	625 CFM @ 80 FPM	12"	EPOXY RESIN FLOOR	HDPE	HDPE	TO BE SELECTED BY COR	SEE BELOW
HOOD LEFT SIDE FACE: 1.(3) 120 V, 20 AMP ELECTRICAL OUTLETS 2.LIGHT SWITCH 3.(3) PASS THROUGH GROMMENTS				HOOD RIGHT SIDE FACE: 1.(1) NITROGEN NEEDLE GAS VALVE 2.(1) HELIUM NEEDLE GAS VALVE 3.DIGITAL HOOD MONITOR								

REVISION	DESCRIPTION	BY	SUPV.	DATE

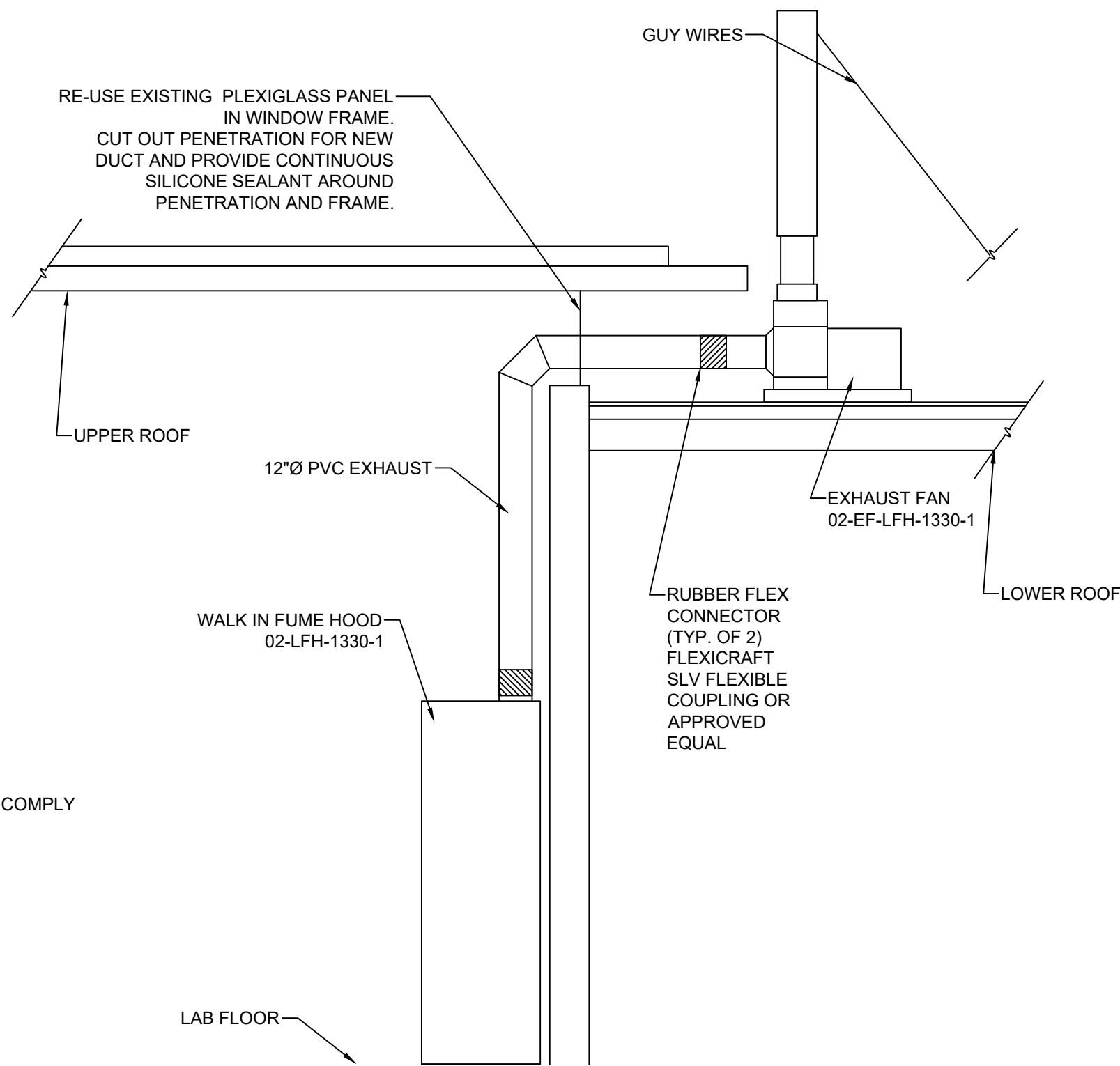
NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY ENGINEERING, MAINTENANCE and SUPPORT SERVICES 325 BROADWAY BOULDER, COLORADO 80305-3328				
BUILDING 2 Room 1330 Walk-In Fume Hood SCHEDULES MECHANICAL				
DESIGN COOK	DRAFTING WIERSMA	APPROVAL	DATE	SHEET NUMBER M-5 of
B2 - Room 1330 Walk-In Fume Hood				



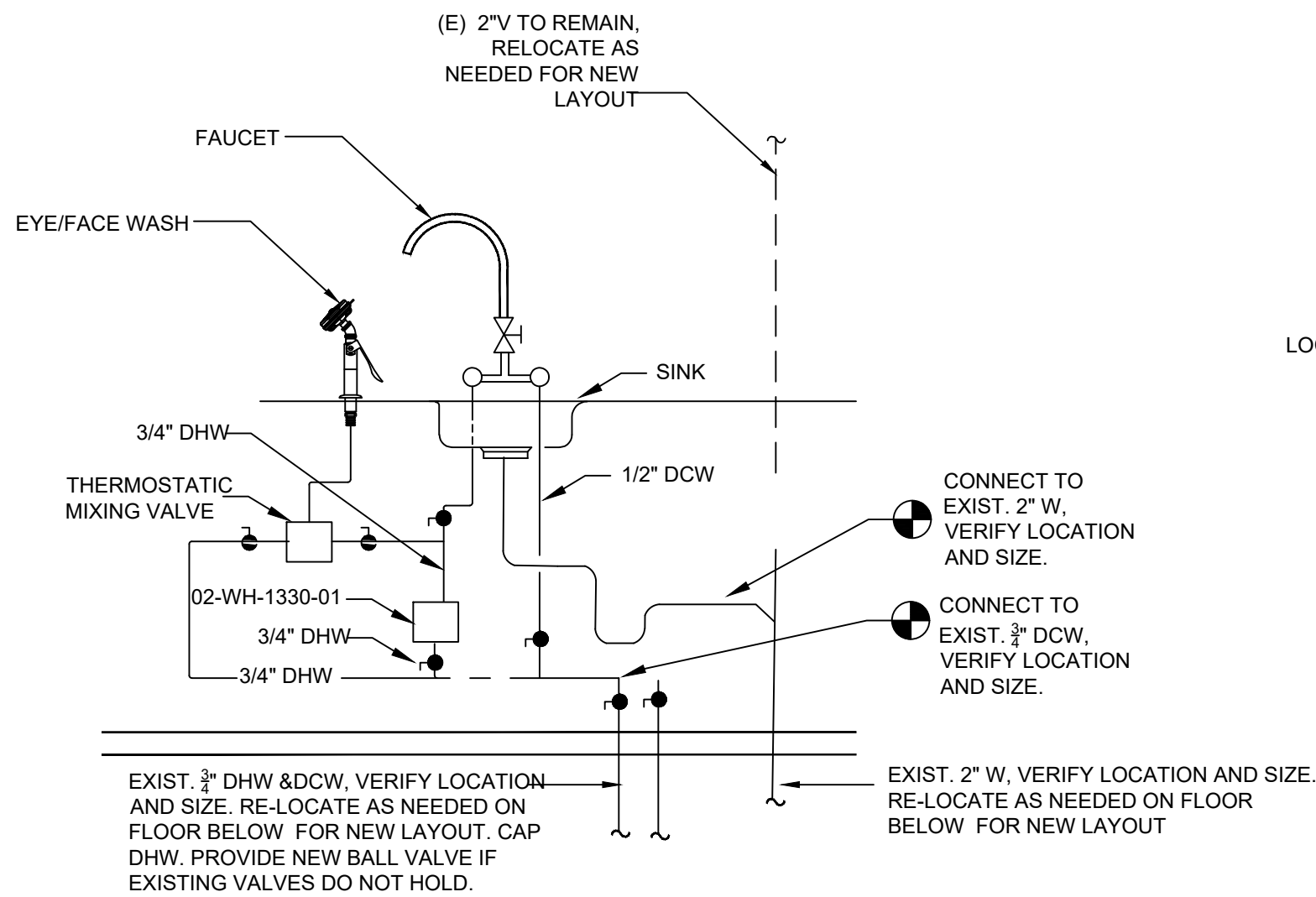
EXHAUST FAN DETAIL
SCALE: NONE



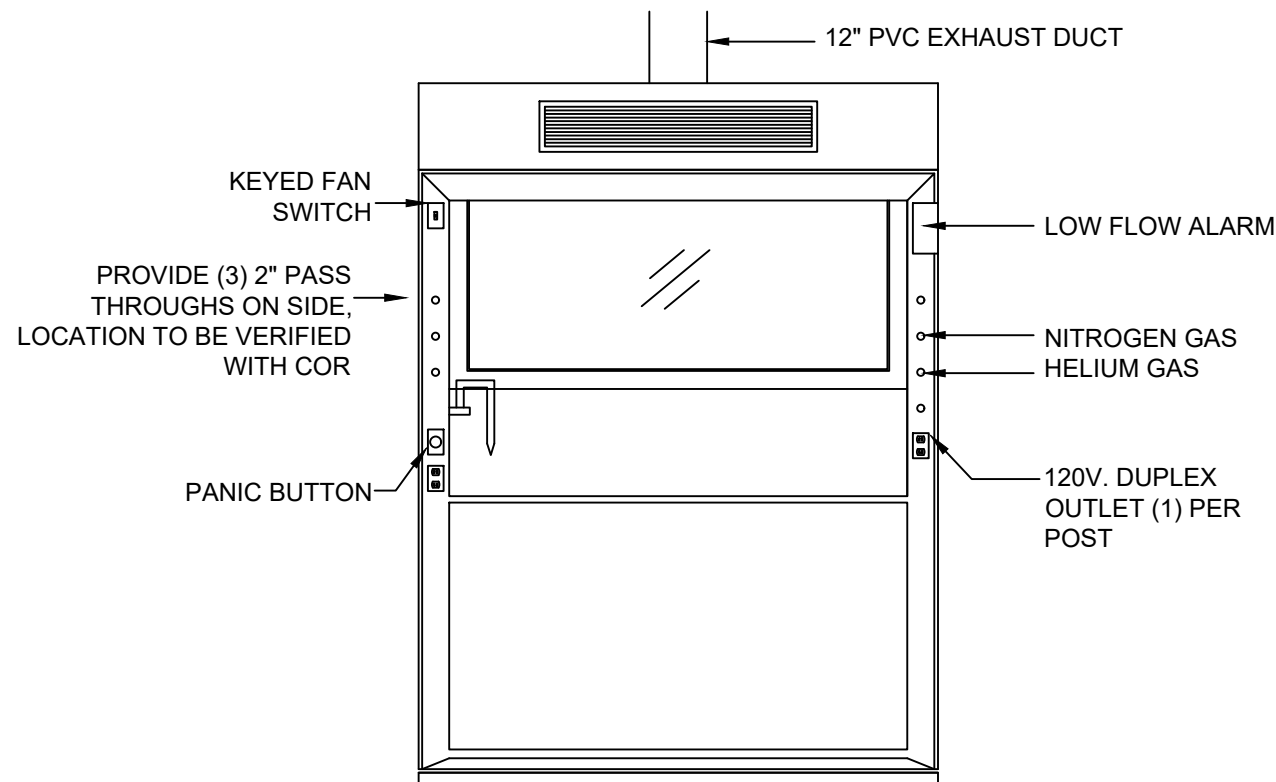
GUY WIRE ANCHOR DETAIL
SCALE: NONE



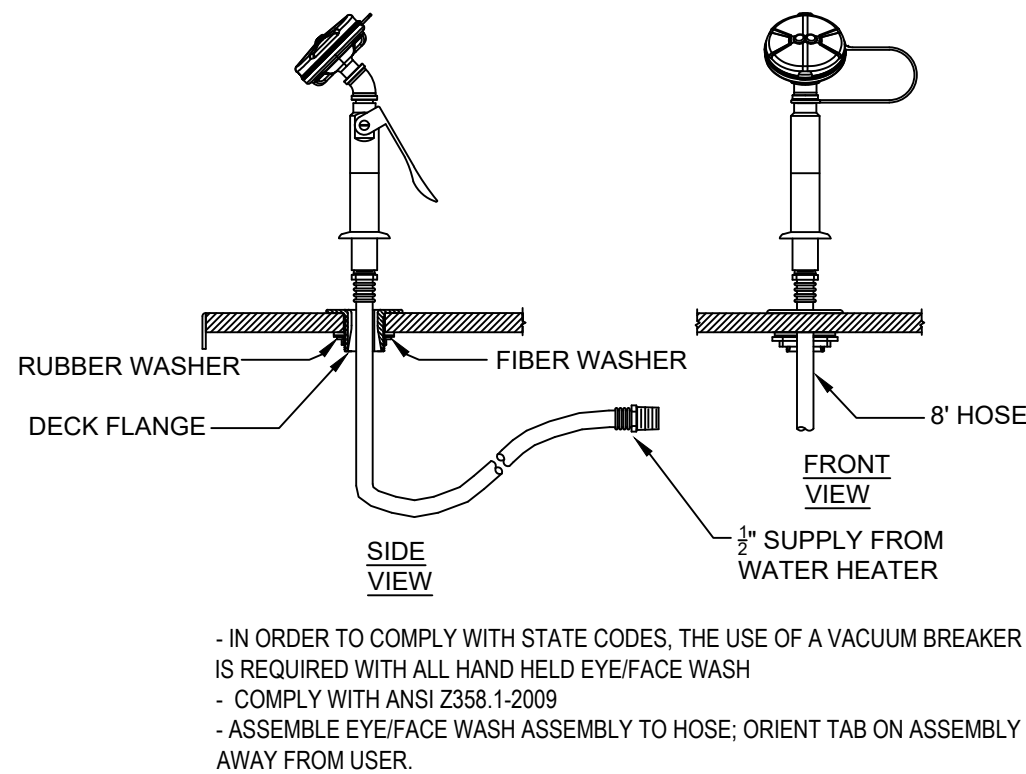
FUME HOOD DETAIL
SCALE: NONE



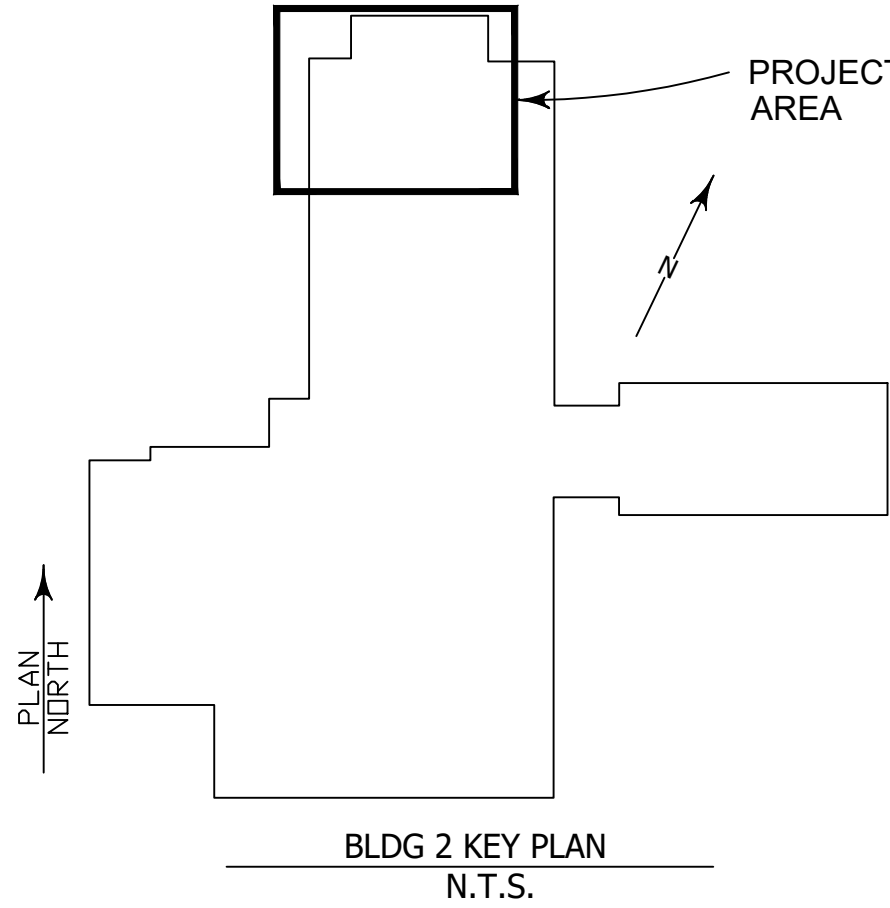
LAB SINK DETAIL
SCALE: NONE



FUME HOOD DETAIL
SCALE: NONE



DECK MOUNT EYE/FACE WASH DETAIL
SCALE: NONE



KEYNOTES
1. KEYNOTES

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY
ENGINEERING, MAINTENANCE and SUPPORT SERVICES
325 BROADWAY BOULDER, COLORADO 80305-3328

BUILDING 2
Room 1330 Walk-In Fume Hood
DETAILS
MECHANICAL

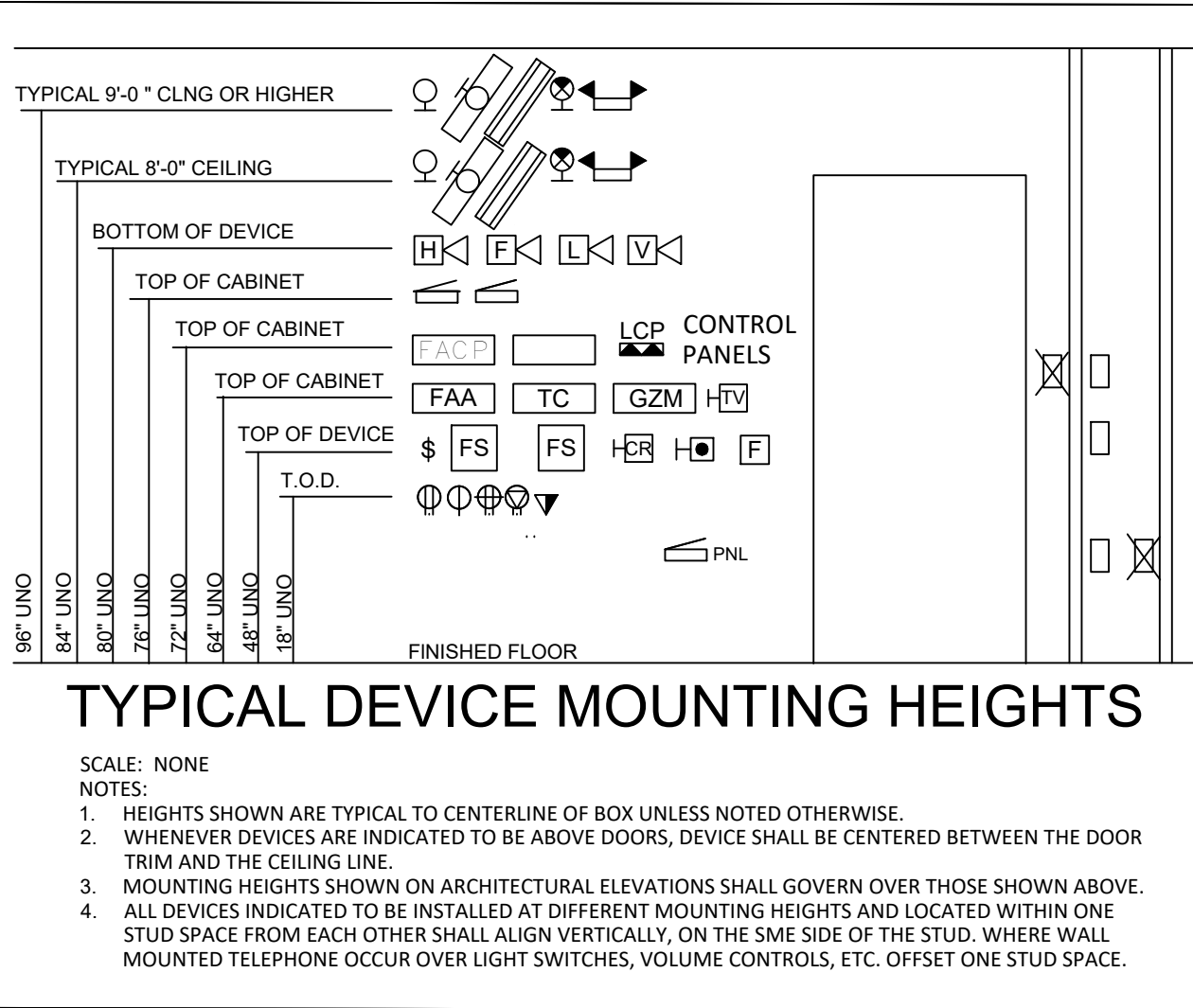
DESIGN	DRAFTING	APPROVAL	DATE	SHEET NUMBER
COOK	WIERSMA			M-6
PROJECT NAME	B2 - Room 1330 Walk-In Fume Hood			
REVISION	DESCRIPTION	BY	SUPV.	DATE

DESIGN	DRAFTING	APPROVAL	DATE	SHEET NUMBER
COOK	WIERSMA			M-6
PROJECT NAME	B2 - Room 1330 Walk-In Fume Hood			
REVISION	DESCRIPTION	BY	SUPV.	DATE

LINE TYPE, CIRCUITING AND HATCHING LEGEND	
SYMBOLS USING THESE LINE WEIGHTS INDICATE PHASING RELATIVE OTHER DEVICES ON SHEET. THIS DOES NOT INDICATE ALL CONSTRUCTION PHASING, CONTRACTOR IS RESPONSIBLE FOR ALL CONSTRUCTION PHASING IN THE PROJECT.	
	DEMOLISH
	EXISTING
	NEW
	FUTURE
	HOME RUN
	CIRCUIT CONTINUATION OR PARTIAL CIRCUIT
	UNDER FLOOR/UNDERGROUND CONDUIT
	LOW VOLTAGE CONDUCTORS
	CONNECTION POINT/EQUIP. TERMINATION
	NIC - NOT IN CONTRACT OR NIS - NOT IN SCOPE
ANNOTATIONS	
	NEW CONNECTION TO EXISTING CIRCUIT
	FLAG NOTE
	USER EQUIPMENT TAG
	UPPER - DETAIL NUMBER
	LOWER - SHEET NUMBER
	REVISION DELTA

ABBREVIATIONS			
AF	AMPERE FUSE SIZE	MLO	MAIN LUGS ONLY
AFC	ABOVE FINISHED CEILING	MLV	MAGNETIC LOW-VOLTAGE
AFF	ABOVE FINISHED FLOOR	MOC	MAXIMUM OVERCURRENT PROTECTION
AFG	ABOVE FINISHED GRADE	MTD	MOUNTED
AHJ	AUTHORITY HAVING JURISDICTION	N/A	NOT APPLICABLE
AHU	AIR HANDLING UNIT	NF	NON-FUSED
AIC	AMPERE INTERRUPTING CAPACITY	NL	NIGHT LIGHT (24HR ON)
AS	AMPERE SWITCH SIZE	NIC	NOT IN CONTRACT
AT	AMPERE TRIP SETTING	NIS	NOT IN SCOPE
ATS	AUTOMATIC TRANSFER SWITCH	NIST	NATIONAL INSTITUTE OF STANDARDS & TECHNOLOGY
AV	AUDIO VISUAL	NRTL	NATIONALLY RECOGNIZED TESTING
BAS	BUILDING AUTOMATION SYSTEM	OPPM	OFFICE OF PROPERTY & MAINTENANCE
BKR	BREAKER	OS	OCCUPANCY SENSOR
C	CONDUIT	OSHE	OFFICE OF SAFETY, HEALTH & ENVIRONMENT
CAT	CATEGORY	P	POLE
CATV	CABLE TELEVISION SYSTEM	PART	PARTIAL CIRCUIT
CCTV	CLOSED CIRCUIT TELEVISION	PH/Ø	PHASE
CD	CANDELA	PNL	PANEL
CKT	CIRCUIT	PNLBD	PANELBOARD
CODE	APPLICABLE CODE ADOPTED BY JURISDICTION	PROVIDE	FURNISH AND INSTALL
CT	CURRENT TRANSFORMER	PT	POTENTIAL TRANSFORMER
CTR	CENTER	QTY	QUANTITY
CTRL	CONTROL/CONTROLLED	R/REL	RELOCATE
CVD	CUMULATIVE VOLTAGE DROP	RCPT	RECEPTACLE
D/DEMO	DEMOLITION	RLA	RUNNING LOAD AMPS
DPDT	DOUBLE-POLE, DOUBLE-THROW	RTU	ROOFTOP UNIT
DPST	DOUBLE-POLE, SINGLE-THROW	SCCR	SHORT-CIRCUIT CURRENT RATING
E/ETR/EX	EXISTING TO REMAIN	SD	SMOKE DUCT DETECTOR
EC	ELECTRICAL CONTRACTOR	SF	SQUARE FEET
EF	EXHAUST FAN	SPDT	SINGLE-POLE, DOUBLE-THROW
EM	EMERGENCY	SPST	SINGLE-POLE, SINGLE-THROW
EMS	ENERGY MANAGEMENT SYSTEM	SSBJ	SUPPLY-SIDE BONDING JUMPER
ELV	ELECTRONIC LOW-VOLTAGE	ST	SHUNT TRIP
ERMS	ENERGY REDUCING MAINTENANCE SWITCH	SWBD	SWITCHBOARD
EWC	ELECTRIC WATER COOLER	SWGR	SWITCHGEAR
FAAP	FIRE ALARM ANNUNCIATOR PANEL	TBB	TELECOMMUNICATIONS BONDING BAR
FACP	FIRE ALARM CONTROL PANEL	TBD	TO BE DETERMINED
FBO	FURNISHED BY OTHERS	TGB	TELECOMMUNICATIONS GROUND BUS BAR
FCA	FAULT CURRENT AMPS AVAILABLE	TL	TWISTLOCK
FCU	FAN COIL UNIT	TMGB	TELECOMMUNICATIONS MAIN GROUND BUS BAR
FF	FINISHED FLOOR	TX/XFMR	TRANSFORMER
FLA	FULL LOAD AMPS	TYP	TYPICAL
FLR	FLOOR	UF	UNDERFLOOR
GC	GENERAL CONTRACTOR	UG	UNDERGROUND
GE	GROUNDING ELECTRODE	US	UNDERSLAB
GES	GROUNDING ELECTRODE SYSTEM	UH	UNIT HEATER
GFR	GROUND FAULT RELAY	UNO	UNLESS NOTED OTHERWISE
G	GROUND	UPS	UNINTERRUPTIBLE POWERSUPPLY
IG	ISOLATED GROUND	VD	VOLTAGE DROP
ISC	SHORT CIRCUIT CURRENT	VFD	VARIABLE FREQUENCY DRIVE
JB/J-BOX	JUNCTION BOX	VS	VACANCY SENSOR
LF	LINEAR FEET	W	WIRE
LRA	LOCKED ROTOR AMPS	W/	WITH
LTG/LTS	LIGHTING/LIGHTS	WP/WR	WEATHER PROOF/WEATHER RESISTANT
MAU	MAKE-UP AIR UNIT	WT	WATERTIGHT
MAX	MAXIMUM	XP	EXPLOSION-PROOF
MCA	MINIMUM CIRCUIT AMPACITY		
MCB	MAIN CIRCUIT BREAKER		
MCC	MOTOR CONTROL CENTER		
MFR	MANUFACTURER		
MIN	MINIMUM		

NOTE: LIST IS NOT EXHAUSTIVE, NOT ALL ABBREVIATIONS SHOWN WILL BE USED



LUMINAIRES AND LIGHTING CONTROLS	
LIGHT FIXTURES	
	SURFACE MOUNTED LENSED FIXTURES
	RECESSED LENSED FIXTURES
	SUSPENDED LENSED FIXTURES
	RECESSED DOWNLIGHTS
	RECESSED DIRECTIONAL DOWNLIGHTS
	LINEAR STRIP FIXTURE
	WALL MOUNTED SCNCE
	CEILING MOUNTED DECORATIVE PENDENT FIXTURE
	BOLLARD
	POLE MOUNTED SITE LIGHTING (UP TO 4 HEADS)
	EMERGENCY LIGHTING UNIT
	CEILING MOUNTED EXIT SIGN (NOT ALL CONFIG. SHOWN)
	WALL MOUNTED EXIT SIGN (NOT ALL CONFIG. SHOWN)
SHADING INDICATES EMERGENCY LIGHT XX = FIXTURE TYPE, REFERENCE LUMINAIRE SCHEDULE x = SWITCH LEG DESIGNATION REFER TO LIGHT FIXTURE SCHEDULE FOR MORE INFORMATION	

LIGHTING CONTROLS	
SWITCH TYPES:	
+\$	+ - SWITCH TYPE DESIGNATION (APPLIES TO ALL DEVICES)
NONE	- SINGLE POLE
2	- DOUBLE POLE
3	- 3-WAY
4	- 4-WAY
D	- LINE VOLTAGE DIMMER
K	- KEY-OPERATED
OSH	- OCCUPANCY/VACANCY SENSOR
L#	- LOW VOLTAGE NON-NETWORKABLE
B#	- LOW VOLTAGE NETWORKABLE
T	- TIME SWITCH
P	- PILOT LIGHT
#	- INDICATES DEVICE INSTANCE REFER TO PLANS FOR MORE INFORMATION
R01	ROOM CONTROLLER (NETWORKABLE, REFER TO SCHEDULES)
PP	POWER PACK (NON-NETWORKABLE, REFER TO SCHEDULES)
PC	PHOTOCELL
OS	CEILING MOUNTED OCCUPANCY SENSOR (REFER TO SCHEDULES)
OS#	CORNER MOUNTED OCCUPANCY SENSOR (REFER TO SCHEDULES)
LCP	LIGHTING CONTROL PANEL
CA	CONTACTOR
CL#	CURRENT LIMITING DEVICE
BTS	BRANCH CIRCUIT TRANSFER SWITCH
ALC	AUTOMATIC LOAD CONTROL RELAY

TELECOMMUNICATIONS	
HTV	IN-WALL TV BOX
W	WALL MOUNTED TELECOMMUNICATIONS OUTLET
W	TELECOMMUNICATIONS FLOORBOX OUTLET
W	TELECOMMUNICATIONS POKE-THRU DEVICE
W	CEILING MOUNTED TELECOMMUNICATIONS OUTLET
WAP	WIRELESS ACCESS POINT
TMGB	TELECOM MAIN GROUND BAR, TELECOM GROUND BAR
TB	TELECOMMUNICATIONS BACKING BOARD

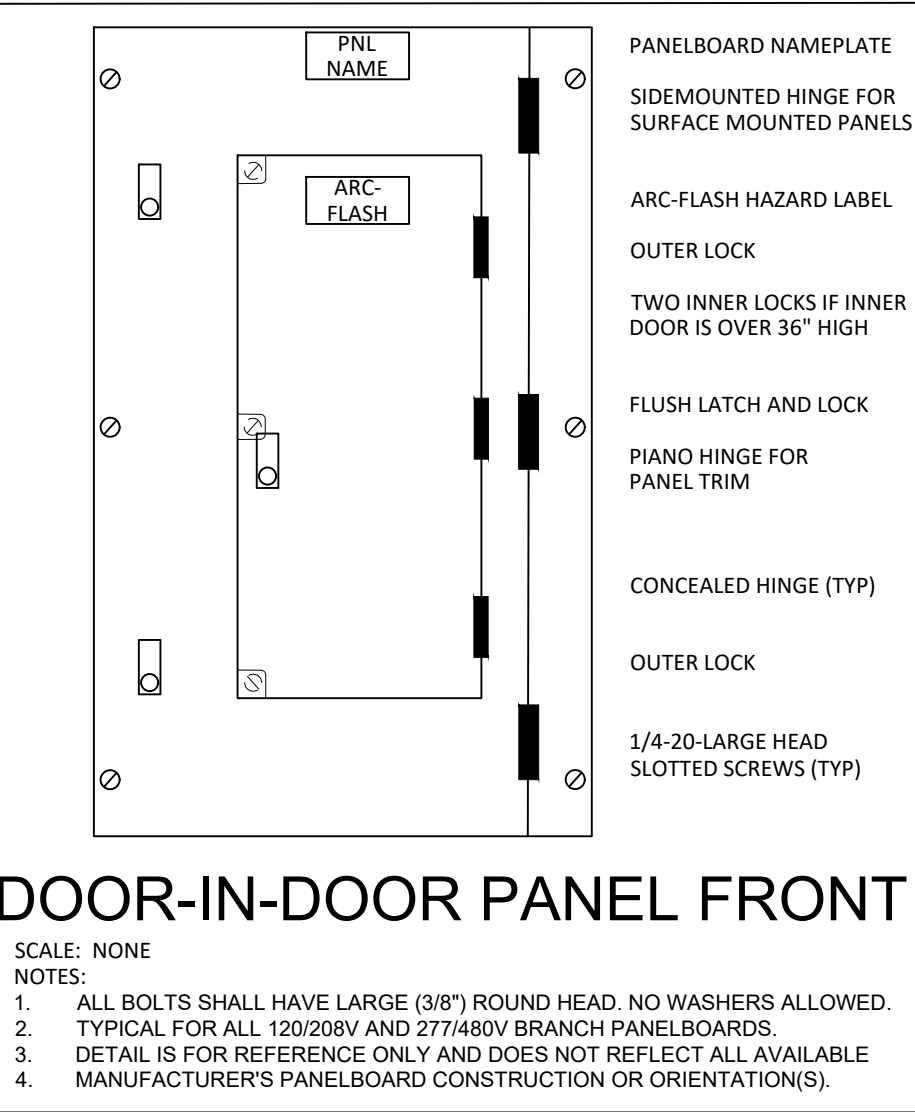
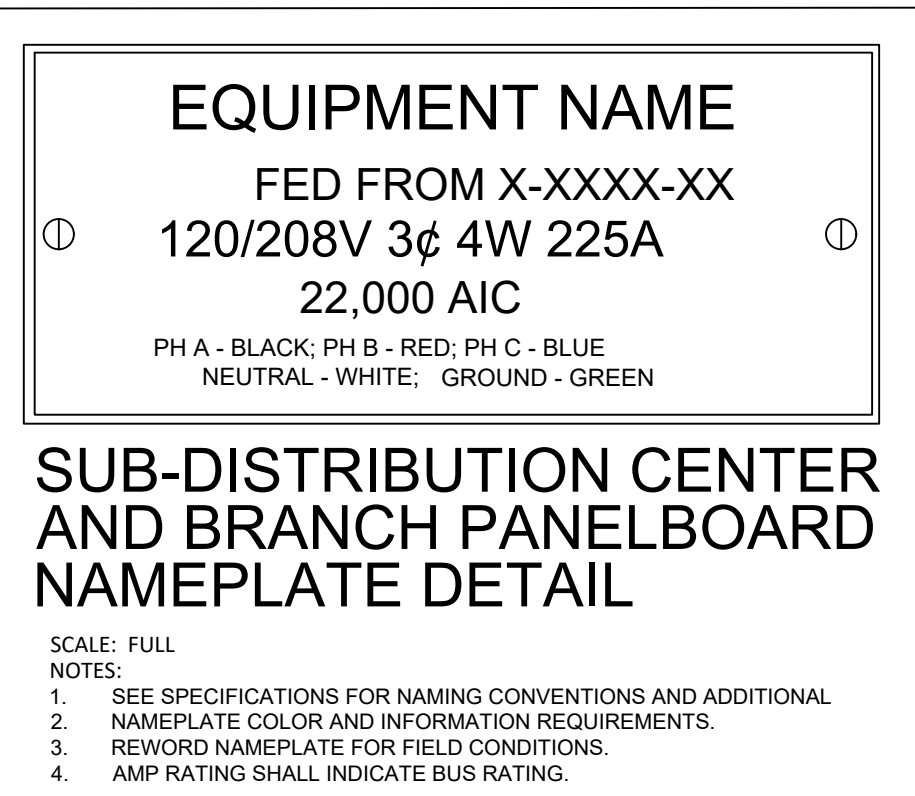
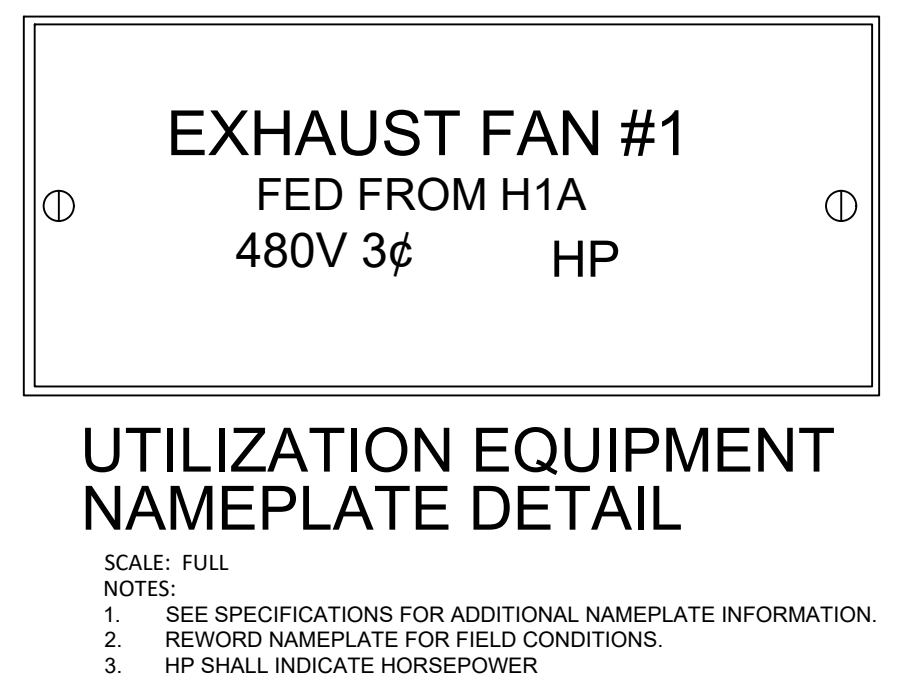
FIRE ALARM DEVICES	
L	LIFE SAFETY SPEAKER, LIFE SAFETY SPEAKER/STROBE
H	HORN, HORN STROBE, STROBE
F	PULL STATION
E	MAGNETIC FIRE DOOR HOLDERS
SP	SMOKE DETECTOR: TYPE DESIGNATIONS ABBREVIATIONS
P	- PHOTOELECTRIC
I	- IONIZATION
T	- COMBINATION
TF	- COMPENSATION
PT	- PHOTOELECTRIC AND THERMAL
TF165	- COMBINATION 165 DEGREES
TF165	- COMBINATION 165 DEGREES
T190	- COMBINATION 190 DEGREES
TF190	- COMBINATION 190 DEGREES
FACP	FIRE ALARM CONTROL PANEL, FIRE ALARM ANNUNCIATOR PANEL
FAAP	

POWER	
RECEPTACLES	
SIMPLEX	ABBREVIATIONS (MAY BE COMBINED)
⊕+	+ - RECEPTACLE TYPE DESIGNATION OR NEMA CONFIGURATION (APPLIES TO ALL DEVICES)
DUPLEX	AC - ABOVE COUNTER
⊕+	C, CH - CONTROLLED
QUADRUPEX	CL - CLOCK HANGAR
⊕+	H - HORIZONTALLY MOUNTED
GENERATOR BACKED	HD - HEAVY DUTY
⊕⊕⊕⊕+	IG - ISOLATED GROUND
CEILING MOUNTED	GF - GFCI PROTECTED OUTLET
⊕⊕⊕⊕⊕	RC - RECESSED
⊕⊕⊕+	SW - SWITCHED
⊕⊕⊕+	TR - TAMPER RESISTANT
⊕⊕⊕+	USB - USB
⊕⊕⊕+	WP - WEATHER RESISTANT, WEATHER PROOF
FLOOR MOUNTED	NOTE: ALL RECEPTACLES TO BE NEMA 5-20R UNO
⊕⊕⊕⊕⊕⊕	POWER ONLY FLOORBOX CONFIGURATIONS
⊕⊕⊕⊕⊕⊕+	POWER AND TELECOM FLOORBOX CONFIGURATIONS
⊕⊕⊕⊕+	POKE THRU DEVICE - REFER TO PLANS FOR CONFIGURATION
⊕⊕⊕	RECESSED CLOCK RECEPTACLE
⊕	POWER POLE
SPECIAL NEMA TYPE RECEPTACLE	
⊕+	- NEMA PLUG TYPE AS SHOWN
WIRING DEVICES	
⊕⊕	JUNCTION BOXES: WALL AND CEILING MOUNTED
WM	SURFACE MOUNT RACEWAY - WIREMOLD 3000 SERIES
⊕⊕	CONDUIT UP, CONDUIT DOWN, CONDUIT STUB OUT
⊕⊕	THERMOSTAT

ELECTRICAL EQUIPMENT	
PNL	SURFACE OR FLUSH MOUNT ELECTRICAL PANEL
PNL	RECESSED ELECTRICAL PANEL
D, PNL	DISTRIBUTION PANELBOARD
SWBD	SWITCHBOARD
MCC	MOTOR CONTROL CENTER
XXX	ELECTRICAL EQUIPMENT
TX-	TRANSFORMER
DISCONNECT ABBREVIATIONS:	
FUSED DISCONNECT	+ DISCONNECT TAG DESIGNATIONS (APPLIES TO ALL DISCONNECTING TYPE DEVICES)
FX/XX/XXX	TO - THERMAL OVERLOAD
NON-FUSED DISCONNECT	FH - FRACTIONAL HORSEPOWER
FX/XX/XXX	FA - FACTORY APPLIED
ENCLOSED CIRCUIT BREAKER	M - MANUAL MOTOR STARTER
FX/XX/XXX	OL - SIZE INDICATED ON ONE-LINE
DISCONNECT	3R - NEMA 3R RATING
FX/XX/XXX	4X - NEMA 4X RATING
	12X - NEMA 12X RATING
	XP - EXPLOSION PROOF
	WP - WEATHER PROOF
FUSED AND NON-FUSED DISCONNECT	
XX/XX/X	XX/XX/X = AMPS/POLES/(FUSE SIZE/TYPE)/ENCLOSURE RATING
ENCLOSED CIRCUIT BREAKER	
XX/X/X	XX/X/X = FRAME/POLES/TRIP/ENCLOSURE RATING

MGB	MAIN GROUNDING BAR, GROUNDING BAR
VFD	VARIABLE FREQUENCY DRIVE
SSRV	STARTER, SOLID STATE REDUCED VOLTAGE STARTER
FS	EMERGENCY POWER OFF PUSH BUTTON, PUSH BUTTON
FS	
FS	HAND ON-OFF AUTO, PUSH BUTTON STOP-START
FS	
FS	INDICATING LIGHT
M	ELECTRICAL METERS
M	
M	

SECURITY & LOW VOLTAGE DEVICES	
CR	CARD READER
GBS	GLASS BREAK SENSOR
PN	PUBLIC NOTIFICATION SPEAKER
C	CAMERA
LPB	LOW VOLTAGE PUSH BUTTON
S	SPEAKER



NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY ENGINEERING, MAINTENANCE AND SUPPORT SERVICES 325 BROADWAY BOULDER, COLORADO 80305-3328				
BUILDING 2 Room 1330 Walk-In Fume Hood				
ELECTRICAL				
DESIGN SJM	DRAFTING SJM	APPROVAL	DATE 01/30/2026	SHEET NUMBER E-0
PROJECT NAME B2 - Room 1330 Walk-In Fume Hood				of

REVISION	DESCRIPTION	BY	SUPV.	DATE
1				
2				
3				
4				

GENERAL TELECOM NOTES

1.

ALL TELECOM AND NETWORK CABLING AND EQUIPMENT SHALL CONFORM WITH AND BE INSTALLED ACCORDING TO OISM/CURRENT NIST NETWORK INFRASTRUCTURE STANDARDS AND ARE SUBJECT TO BUY AMERICAN COMPLIANCE ACT.
2.

MINIMUM TELECOM CONDUIT SIZE IS 1" ALL CONDUITS TO BE ROUTED TO ACCESSIBLE SPACE AND PROVIDED WITH PLASTIC BUSHING AND PULL STRINGS.
3.

TELECOM JACK REQUIREMENTS:
 - TELECOM OUTLETS SHALL HAVE ONE VOICE CABLE/JACK AND THREE DATA CABLES/JACKS PER USER STATION.
 - BLUE KEYSTONE JACKS SHALL BE USED FOR DATA JACKS.
 - WHITE KEYSTONE JACKS SHALL BE USED FOR TELEPHONE JACKS.
 - ALL CABLES SHALL BE COMMSCOPE CERTIFIED PRIOR TO PROJECT CLOSEOUT.
 - NO NIST SYSTEMS SHALL BE CONNECTED TO NEW CABLE INSTALLATIONS WITHOUT COMMSCOPE CERTIFICATION AND 25 YEAR WARRANTY.
3.

CABLE REQUIREMENTS:
 - DATA CABLES SHALL BE BLUE.
 - PHONE CABLES SHALL BE WHITE.
4.

CABLE AND JACK LABELLING:
 - ALL PATCH PANEL AND WALL JACKS SHALL BE LABELED PER NIST SITE STANDARD. COORDINATE WITH NIST ON LABELING NOMENCLATURE.
 - PATCH PANELS AND WALL JACKS SHALL BE LABELED SEQUENTIALLY BASED ON ROOM NUMBER/STATION NUMBER/JACK LETTER.
 - JACKS STATION NUMBERS SHALL BE LABELED AND INCREMENTED IN A COUNTERCLOCKWISE FASHION FROM ROOM ENTRY STARTING FROM THE LEFT.
 - LABELS SHALL BE SNAP-IN, CLEAR-LABEL COVERS AND MACHINE-PRINTED PAPER INSERTS.
5.

CONTRACTOR REQUIREMENTS AND RESPONSIBILITIES:

i.

COMMSCOPE CERTIFICATION
 - THE CONTRACTOR PROVIDING THE CERTIFICATION SHALL BE A COMMSCOPE CERTIFIED COMPANY WHO IS AUTHORIZED BY THE REGIONAL COMMSCOPE AUTHORITY AND SHALL BE AUTHORIZED IN BOULDER OR GAITHERSBURG REGIONAL AREAS.

ii.

KEY PERSONNEL TECHNICAL QUALIFICATIONS
 - PROJECT MANAGER(S):
 - SHALL HAVE AT LEAST 3 YEARS' EXPERIENCE ESTIMATING AND INSTALLING COMMSCOPE SYSTIMAX FIBER AND COPPER CABLING PRODUCTS ON JOBS WHERE COMMSCOPE CERTIFICATION IS REQUIRED.
 - SHALL HAVE PROJECT MANAGEMENT PROFESSIONAL (PMP) CERTIFICATION.
 - SHALL HAVE REGISTERED COMMUNICATIONS DISTRIBUTION DESIGNER (RCDD) CERTIFICATION.
 - SHALL BE CERTIFIED IN COMMSCOPE DESIGN AND ENGINEERING COURSE (SP3321)
 - FIBER TECHNICIANS:
 - SHALL HAVE AT LEAST 3 YEARS' EXPERIENCE INSTALLING COMMSCOPE SYSTIMAX FIBER CABLING PRODUCTS ON JOBS WHERE COMMSCOPE CERTIFICATION IS REQUIRED.
 - SHALL BE CERTIFIED IN COMMSCOPE FIBER OPTICS SP4400 AND SP4410 COURSES.
 - CAT6/CAT6A TECHNICIANS:
 - SHALL HAVE AT LEAST 2 YEARS' EXPERIENCE INSTALLING COMMSCOPE SYSTIMAX CAT6 CABLING PRODUCTS ON JOBS WHERE COMMSCOPE CERTIFICATION IS REQUIRED.
 - SHALL BE CERTIFIED IN COMMSCOPE INSTALLATION AND MAINTENANCE ND3361 OR FC3361 COURSES.
 - SHALL BE BICSI CERTIFIED.
 - RESUMES OF CONTRACTOR KEY PERSONNEL SHALL BE PROVIDED IN PROPOSAL.

6.

COPPER INSTALLATION
 - TERMINATION AND TESTING OF CAT6/6A SHALL BE DONE BY COMMSCOPE CERTIFIED CAT6/6A INSTALLERS.
 - ALL CAT6/6A CABLE SHALL BE PLENUM RATED UNSHIELDED COMMSCOPE SYSTIMAX CABLE UNLESS ANOTHER CABLE STANDARD IS REQUESTED BY USERS.
 - ALL DATA CABLES ARE TO BE BLUE CATEGORY 6/6A.
 - PATCH PANELS AND CABLES SHALL BE CLEARLY LABELED WITH A MACHINE PRINTED LABEL ACCORDING TO THE LABELING SPECIFICATION GIVEN ON THE CUTSHEET PROVIDED BY THE COR.
 - ELECTRONIC COPY OF END-TO-END POWER METER TRACES AS REQUIRED BY COMMSCOPE FOR SUBMISSION FOR WARRANTY. PROVIDE COR AND NETWORK OPERATIONS TEAM WITH PDF FORMAT.
 - WHERE ANY COPPER CABLE DOES NOT MEET COMMSCOPE SPECIFICATIONS FOR WARRANTY THE CABLE SHALL BE RE-TERMINATED AND RE-TESTED AT NO ADDITIONAL COST TO NIST.
 - USE CABLE SUPPORT BARS TO ELIMINATE SLACK AT CABLE RACK. CABLE SHOULD NOT HANG BELOW THE RACK UNIT IN WHICH IT IS INSTALLED. CABLE SLACK CANNOT IMPEDE EQUIPMENT PLACEMENT.
 - FIVE FEET OF CABLE SLACK FOR STATION CABLE SHALL BE PLACED NEAR END USER STATION.
 - CUSTOMER REQUIREMENTS WILL GENERATE ACTUAL CABLE DENSITY REQUIREMENTS.
 - DURING RENOVATIONS OLD COPPER CABLING SHALL BE REMOVED FROM STATION TO MDF/IDF.

7.

PROJECT SEQUENCING
 - ALL COMMUNICATION LINES SHALL BE TESTED AND CERTIFIED PRIOR TO ACCEPTANCE BY THE NETWORK OPERATIONS TEAM
 - NO NETWORK EQUIPMENT OR ENDPOINTS SHALL BE CONNECTED UNTIL THE COMMSCOPE 25-WARRANTY CERTIFICATION RECEIVED.
 - PROJECT MILESTONES SHOULD BE ESTABLISHED TO CONSULT WITH NETWORK TEAM TO INSPECT WORK

GENERAL POWER NOTES

1.

ALL CIRCUITY SHALL BE #12 AWG IN 1/2" CONDUIT, MINIMUM, UNLESS OTHERWISE NOTED.

2.

PROVIDE A SEPARATE CODE SIZED GREEN EQUIPMENT GROUNDING CONDUCTOR IN ALL CONDUITS AND RACEWAYS CONTAINING LINE VOLTAGE CIRCUITS. FOR ALL 20A CIRCUITS, EQUIPMENT GROUNDING CONDUCTOR SHALL MATCH PHASE CONDUCTOR SIZE. FOR CIRCUITS UPSIZED DUE TO VOLTAGE DROP, INCREASE EQUIPMENT GROUNDING CONDUCTOR SIZE PER NEC 2017 250.122.B.

3.

PROVIDE A MINIMUM OF FOUR (4) SPARE 1" CONDUITS FROM RECESSED PANELBOARDS, UP TO ACCESSIBLE CEILING SPACE.

4.

PROVIDE A NEUTRAL CONDUCTOR TO THREE-PHASE EQUIPMENT WHEREVER REQUIRED BY THE EQUIPMENT MANUFACTURER.

5.

ALL SINGLE PHASE 120V, 208V OR 240V RECEPTACLES 50A OR LESS, AND THREE PHASE 208V AND 240V RECEPTACLES 100A OR LESS, SHALL BE GFCI PROTECTED IN LOCATIONS REQUIRED BY CODE; THIS INCLUDES BATHROOMS, KITCHENS/FOOD PREP AREAS, DRINK PREP AREAS, EXTERIOR LOCATIONS AND RECEPTACLES WITHIN 6 FEET OF A SINK. GFCI RECEPTACLES SHALL BE READILY ACCESSIBLE AND SHALL NOT BE LOCATED BEHIND STATIONARY EQUIPMENT. GFCI PROTECTION MAY BE VIA A GFCI CIRCUIT BREAKER OR GFCI RECEPTACLE, UNLESS NOTED OTHERWISE WHERE NECESSARY, GFCI PROTECTION MAY BE ACHIEVED VIA A BLANK FACE GFCI DEVICE LOCATED IN A READILY ACCESSIBLE LOCATION NEAR RECEPTACLE BEING PROTECTED. FOR DOWNSTREAM WIRING DEVICES LOCATED ON THE SAME BRANCH CIRCUIT, THE GFCI PROTECTION MAY BE PROVIDED FOR BY A SINGLE UPSTREAM DEVICE IF ALL PROTECTED DEVICES ARE LABELED PER CODE.

6.

GROUND AND NEUTRAL CONDUCTORS SHALL NOT BE SHARED UNLESS SPECIFICALLY NOTED ON PLANS.

7.

ALL GFCI PROTECTED CIRCUITS SHALL HAVE INDIVIDUAL AND DEDICATED NEUTRALS

8.

VERIFY REQUIREMENTS OF ALL MECHANICAL EQUIPMENT WITH SHOP DRAWING SUBMITTALS. NOTIFY ENGINEER OF ANY CONFLICTS BETWEEN SUBMITTALS AND ELECTRICAL DRAWINGS.

9.

RACEWAYS INSTALLED ON ROOFTOPS OR EXPOSED TO DIRECT SUNLIGHT CORRECTIONS NEED TO BE PROVIDED FOR CONDUCTOR AMPACITY AND OR SIZING BASED ON AMBIENT TEMPERATURE CORRECTION FACTORS. TEMPERATURE CORRECTION FACTORS SHOWN IN NEC TABLE 310.15(B)(3)(C) SHALL BE ADDED TO THE OUTDOOR TEMPERATURE TO DETERMINE THE APPLICABLE AMBIENT TEMPERATURE FOR APPLICATION OF THE CORRECTION FACTORS IN TABLE 310.15(B)(2)(A) OR TABLE 310.15(B)(2)(B).

10.

AN INDIVIDUAL DISCONNECTING MEANS SHALL BE PROVIDED LOCATED IN SIGHT FROM EACH MOTOR OR DRIVEN MACHINERY IN ACCORDANCE WITH NEC ARTICLE 430.

10.

MAINTAIN DEDICATED EQUIPMENT AND WORKING SPACE IN FRONT AND ABOVE ALL ELECTRICAL EQUIPMENT REQUIRING SERVICING WHILE ENERGIZED PER NEC 110.26. THIS INCLUDES CONTROL PANELS AND ELECTRICAL DISCONNECTS FOR HVAC EQUIPMENT ON LOCATED ON ROOFTOPS AND ABOVE/BELOW CEILING. PENETRATIONS SUCH AS ROOF JACKS FOR ELECTRICAL POWER, LOW VOLTAGE CONTROL POWER, REFRIGERANT LINES, VENT PIPES, ETC., AND INCLUDING GAS LINES, DUCTWORK, ROOF DRAINS, SCREENING WALLS AND OTHER EQUIPMENT OF ANY TYPE, ARE NOT TO INTRUDE INTO DEDICATED ELECTRICAL EQUIPMENT SPACE.

GENERAL NOTES

1.

PRIOR TO SUBMITTING BID, VISIT THE JOB SITE AND BECOME FULLY ACQUAINTED WITH THE EXISTING CONDITIONS. REVIEW OPM/NIST PROJECT CRITERIA, GENERAL NOTES, OTHER TRADE DRAWINGS AND OPM/NIST SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS THAT MAY NOT BE CALLED OUT IN THIS PORTION OF THE CONSTRUCTION DOCUMENTS. READ THE SPECIFICATIONS AND REVIEW DRAWINGS OF ALL DIVISIONS OF WORK. COORDINATE THIS WORK WITH ALL OTHER DIVISIONS OF WORK AND ALL SUBCONTRACTORS. PROVIDE ALL SUBCONTRACTORS WITH A COMPLETE SET OF BID DOCUMENTS. NOTIFY ARCHITECT AND ENGINEER OF ANY CONFLICTS OR DISCREPANCIES PRIOR TO SUBMITTING BID.

2.

DRAWINGS ARE DIAGRAMMATIC ONLY AND REPRESENT THE GENERAL SCOPE OF WORK. REVIEW THE GENERAL NOTES, SPECIFICATIONS, AND PLANS FOR ADDITIONAL REQUIREMENTS THAT MAY NOT BE SPECIFICALLY NOTED IN THIS PORTION OF THE CONSTRUCTION DOCUMENTS. IT IS NOT WITHIN THE SCOPE OF THE ELECTRICAL DRAWINGS TO SHOW ALL NECESSARY RACEWAY ROUTING, BENDS, OFFSETS, PULL BOXES AND OBSTRUCTIONS. THE DRAWINGS REPRESENT THE BEST INFORMATION AVAILABLE TO THE ENGINEER. ALL DIMENSIONS AND SIZES SHALL BE FIELD VERIFIED. DO NOT SCALE FROM THE DRAWINGS. SMALL DEVIATIONS SHALL BE RECONCILED DURING THE PERFORMANCE OF THE WORK. NOTIFY THE ARCHITECT OF ANY CONFLICTS OR DISCREPANCIES PRIOR TO SUBMISSION OF BID.

3.

ALL WORK SHALL CONFORM TO OPM/NIST STANDARDS, ALL NATIONAL CODES AS WELL AS APPLICABLE INDUSTRY STANDARDS. ALL EQUIPMENT SHALL BEAR LABELS FOR THE USE INTENDED BY AN ACCEPTED NATIONALLY RECOGNIZED TESTING LABORATORY(NRTL), SUCH AS UL OR ETL. THE FINAL ELECTRICAL INSTALLATION OF THE FACILITY SHALL BE FREE FROM ELECTRICAL DEFECTS TO THE SATISFACTION OF OPM/NIST PERSONNEL.

4.

COORDINATE FINAL LOCATION AND INSTALLATION REQUIREMENTS OF ALL WORK: LIGHT FIXTURES, ELECTRICAL EQUIPMENT AND ELECTRICAL DEVICES, ETC. WITH ARCHITECTURAL DRAWINGS, EXISTING CONDITIONS AND OTHER TRADES AND CONTRACTORS PRIOR TO ROUGH-IN. PROVIDE ALL NECESSARY DEVICES, CORDS, PLUGS, DISCONNECTS AND FINAL CONNECTIONS TO ELECTRICAL EQUIPMENT FOR PROPER OPERATION IN ACCORDANCE WITH CODE, OWNER AND MANUFACTURER REQUIREMENTS

5.

REFER TO OPM/NIST STANDARDS DOCUMENTS: CONDUIT AND LABEL COLOR CHART FOR COLOR CODING CONDUITS AND LABELING OF SYSTEMS BY SYSTEM TYPE. ELECTRICAL STANDARDS FOR SYSTEM PERFORMANCE CRITERIA. EQUIPMENT LIST AND IDS FOR EQUIPMENT LABEL IDENTIFICATION.

6.

DRAWINGS AND SPECIFICATIONS GOVERN, WHERE THEY EXCEED CODE REQUIREMENTS. SYSTEM INSTALLATION SHALL COMPLY WITH LEGALLY CONSTITUTED CODES AND THE REQUIREMENTS OF NIST/OPFM.

7.

FIELD VERIFY EXACT LOCATIONS AND ELECTRICAL REQUIREMENTS OF ALL HVAC AND PLUMBING EQUIPMENT WITH OTHER TRADE CONTRACTORS PRIOR TO ORDERING RELATED ELECTRICAL EQUIPMENT.

8.

ROOM NAMES/NUMBERS SHOWN IN PANELBOARD SCHEDULES ARE PER ARCHITECTURAL FLOOR PLANS. CONTRACTOR SHALL PROVIDE FINALIZED PANELBOARD SCHEDULES AT COMPLETION OF PROJECT WITH NIST PROVIDED ROOM NAMES/NUMBERS. SEE SPECIFICATIONS FOR ADDITIONAL INFORMATION. PROVIDE UPDATED TYPED PANEL DIRECTORIES TO REFLECT WORK PERFORMED UNDER THIS CONTRACT.

9.

PROVIDE A CONSTRUCTION RECORD SET OF "AS-BUILT" DOCUMENTS REFLECTING ANY VARIANCES OF INSTALLED PIPING, EQUIPMENT DEVICES, ETC LOCATIONS THAT ARE CONTRARY TO THE CONSTRUCTION DOCUMENTS. REFER TO SPECIFICATIONS FOR ADDITIONAL INFORMATION.

10.

WHERE DEVICES ARE MOUNTED RECESSED IN CMU WALLS, ROUTE CONDUIT CONCEALED WITHIN INTERIOR OPENINGS WITHIN CMU WALL. COORDINATE WITH GENERAL CONTRACTOR PRIOR TO ROUGH-IN.

11.

ALL DEVICES SHOWN DIRECTLY ADJACENT TO EACH OTHER SHALL BE INSTALLED DIRECTLY ADJACENT TO EACH OTHER, U.N.O. ADJACENT DEVICES OF SIMILAR TYPE SHALL BE PROVIDED WITH SINGLE FACEPLATE WHERE FEASIBLE.

12.

ALL APPLICABLE SWITCHES, RECEPTACLES, OUTLETS, AND CONTROLS SHALL BE PLACED AT HEIGHTS THAT ARE IN ACCORDANCE WITH ADA ACCESSIBILITY GUIDELINES.

13.

ALL JUNCTION BOXES SHALL BE RIGIDLY ATTACHED TO STRUCTURE OR MILLWORK.

14.

ALL WP OUTLET BOX HOODS SHALL BE "EXTRA-DUTY" AND "WHILE-IN-USE COVER" TYPE. OUTLET BOX HOODS SHALL BE LOW PROFILE WHEREVER PRACTICABLE, UNLESS NOTED OTHERWISE. THE USE OF LARGE BUBBLE COVERS SHALL BE AVOIDED ON THE EXTERIOR OF THE BUILDING OR BEHIND EQUIPMENT IN ORDER TO PREVENT DAMAGE TO THE COVER AND TO ALLOW THE EQUIPMENT TO BE LOCATED CLOSE TO THE WALL.

15.

REFER TO ARCHITECTURAL PLANS AND DETAILS FOR EXACT LOCATIONS, ALIGNMENT, AND MOUNTING OF ALL CEILING, WALL, AND FLOOR MOUNTED DEVICES.

16.

CONDUIT AND BOX ROUGH-IN FOR ADJACENT FIRE ALARM, TEMPERATURE CONTROLS, RECEPTACLES, LIGHTING CONTROL DEVICES, ETC., ON THE SAME WALL, SHALL BE PROVIDED IN A MANNER TO WHICH DEVICES ALIGN VERTICALLY ON SAME WALL. COORDINATE WITH OTHER TRADES.

17.

REFER TO TELECOM DRAWINGS FOR BOXES, FLOOR BOXES, RACEWAYS, PATHWAYS, AND CABLE TRAY TO BE BID AS PART OF THE MEPP/FP PACKAGE. REFER TO SPECIFICATION 260510 FOR ADDITIONAL INFORMATION.

18.

COORDINATE CONDUIT PENETRATIONS THROUGH SLABS WITH STRUCTURAL DOCUMENTS. CONDUITS ARE NOT TO BE ROUTED WITHIN A COMPOSITE SLAB. X-RAY CONCRETE PRIOR TO CORE DRILLING AND/OR SAW CUTTING

19.

WHERE SPARE CONDUIT(S) ARE INDICATED FOR FUTURE USE, PROVIDE PULL STRINGS IN CONDUIT(S) AND PROTECTIVE BUSHINGS AT OPENINGS. CAP CONDUITS WHERE LOCATED BELOW GRADE OR EXPOSED TO THE ELEMENTS.

20.

REFER TO MECHANICAL EQUIPMENT SCHEDULES FOR ELECTRICAL SCOPE OF WORK, IN ADDITION TO WORK SHOWN ON ELECTRICAL DRAWINGS.

21.

PROVIDE RECESSED CONDUIT AND OUTLET BOXES FOR ALL HVAC CONTROL LOCATIONS. PROVIDE CONDUIT FOR ALL CONTROLS WIRING LOCATED IN SPACES WITH EXPOSED CEILINGS. REFER TO MECHANICAL DOCUMENTS AND SCHEDULES FOR ALL DEVICE LOCATIONS. COORDINATE WORK WITH MECHANICAL CONTRACTOR.

22.

PROVIDE ALL MISCELLANEOUS STEEL REQUIRED FOR THE PROPER INSTALLATION OF ELECTRICAL EQUIPMENT AND SYSTEMS.

23.

FOR NEW WORK IN EXISTING BUILDINGS POWER AND TELECOM OUTLETS TO BE PROVIDED IN SURFACE MOUNT RACEWAYS, WIREMOLD 3000 SERIES.

24.

REFER TO SPECIFICATIONS AND DETAILS FOR APPROVED CABLE AND RACEWAY INSTALLATION. NON-COMPLIANT INSTALLATIONS OF CABLE AND RACEWAY WILL NOT BE ACCEPTED AND WILL BE REQUIRED TO BE BROUGHT TO COMPLIANCE AT NO COST TO THE OWNER, PRIOR TO COMPLETION OF WORK.

25.

ALL NEW AND EXISTING ELECTRICAL EQUIPMENT ALTERED UNDER THIS PROJECT SHALL BE VACUUM CLEANED OF ANY DEBRIS. ALL OPENINGS REMAINING SHALL BE SEALED WITH THE PROPER DEVICE (IE. KNOCKOUT BLANKS, BREAKER BLANKS, ETC) LISTED AND APPROVED FOR USE.

26.

NO WORK SHALL BE PERFORMED PRIOR TO REVIEW AND APPROVAL OF ALL REQUIRED SHOP DRAWINGS, PRODUCT MATERIALS, AND EQUIPMENT SUBMITTALS. ANY WORK INSTALLED PRIOR TO MEETING THESE REQUIREMENTS SHALL BE DONE SO AT THE SOLE RISK OF THIS CONTRACTOR.

27.

PROVIDE PROPER FIRE PROOFING AND SEALANT FOR PENETRATIONS THROUGH FIRE RATED ASSEMBLIES. THE FIRE STOPPING METHOD, MATERIAL AND ITS APPLICATION SHALL BE NRTL LISTED, CODE COMPLIANT AND APPROVED BY OPM/OSHE/NIST.

GENERAL FIRE ALARM NOTES

1.

CONNECTIONS TO FIRE ALARM SYSTEM TO BE DESIGN BUILD BY QUALIFIED LICENSED FIRE ALARM CONTRACTOR. FIRE ALARM CONTRACTOR TO SUBMIT SIGNED AND STAMPED DRAWINGS OF FIRE ALARM SYSTEM AS A SUBMITTAL TO NIST LIFE SAFETY OFFICE PRIOR TO BEGINNING ANY WORK.

2.

PROVIDE UL RATED FIRE STOPPING ASSEMBLIES AT ALL PENETRATIONS OF FIRE RATED CONSTRUCTION. SEAL ALL PENETRATIONS OF SMOKE WALLS SMOKE TIGHT.

3.

FIRE ALARM CONTRACTOR SHALL PROVIDE ALL WORK (HARDWARE, SOFTWARE, PROGRAMMING, WIRE, CONDUIT, ETC.) NECESSARY TO PROVIDE A COMPLETE AND CODE COMPLIANT SYSTEM.

4.

REMOVE ALL FIRE ALARM COMPONENTS, INCLUDING WIRE AND CONDUIT MADE OBSOLETE BY WORK PERFORMED UNDER THIS CONTRACT. PROVIDE NECESSARY SOFTWARE AND PROGRAMMING REVISIONS.

5.

ALL FIRE DETECTION SYSTEM WIRE TO BE UL PLENUM RATED PER NIST STANDARDS. WIRE TO BE INSTALLED IN RACEWAYS PER NIST STANDARDS .

6.

LABEL ALL FIRE DETECTION SYSTEM WIRE AND RACEWAY PER THE SPECIFICATIONS AND TO MATCH EXISTING FACILITY IDENTIFICATION SYSTEM.

GENERAL DEMOLITION NOTES

1.

THE ELECTRICAL DRAWINGS INDICATE EXISTING ELECTRICAL ITEMS TO BE REMOVED. THE DRAWINGS ARE INTENDED TO INDICATE THE SCOPE OF WORK REQUIRED AND DO NOT INDICATED EVERY BOX, CONDUIT, OR WIRE THAT MUST BE REMOVED. CONTRACTOR SHALL INSPECT THE SITE PRIOR TO THE SUBMISSION OF A BID. CONTRACTOR SHALL INFORM THEMSELVES OF THE CONDITIONS UNDER WHICH THE WORK IS TO BE PERFORMED CONCERNING THE SITE OF THE WORK, THE OBSTACLES WHICH MAY BE ENCOUNTERED, THE DEMOLITION, AND TEMPORARY REMOVAL AND REINSTALLATION REQUIRED TO PROVIDE ACCESS TO THE WORK, AND ALL OTHER RELEVANT MATTERS CONCERNING THE WORK TO BE PERFORMED.

2.

CONTRACTOR SHALL REPAIR ALL DAMAGE TO EXISTING BUILDING, FIXTURES, AND FINISHED CAUSED BY CONTRACTOR DURING THE PERFORMANCE OF THE WORK. REPAIRS SHALL BE PERFORMED BY A QUALIFIED TRADESMEN AND SHALL BE COMPLETED IN A MANNER ACCEPTABLE TO THE OWNER.

3.

REMOVAL OF CONDUITS SHALL INCLUDE REMOVAL OF HANGERS, SUPPORTS, AND ASSOCIATED MISCELLANEOUS MATERIALS.

4.

ALL PIPING, TUBING, CONDUITS, ETC. MADE OBSOLETE, BY WORK UNDER THIS CONTRACT, EXPOSED OR IN CONFLICT WITH NEW WORK, ARE TO BE REMOVED. REPAIR ALL HOLES IN WALLS, FLOORS, AND CEILING TO MATCH EXISTING CONDITIONS AND MAINTAIN FIRE/SMOKE RATINGS.

5.

IT SHALL BE THE RESPONSIBILITY OF THIS CONTRACTOR TO PERFORM ALL SELECTIVE DEMOLITION NECESSARY TO PERFORM THE WORK SHOWN ON THE DRAWINGS EXCEPT WHERE SAID DEMOLITION IS SHOWN ON THE ARCHITECTURAL DRAWINGS TO BE PERFORMED BY THE GENERAL CONTRACTOR.

6.

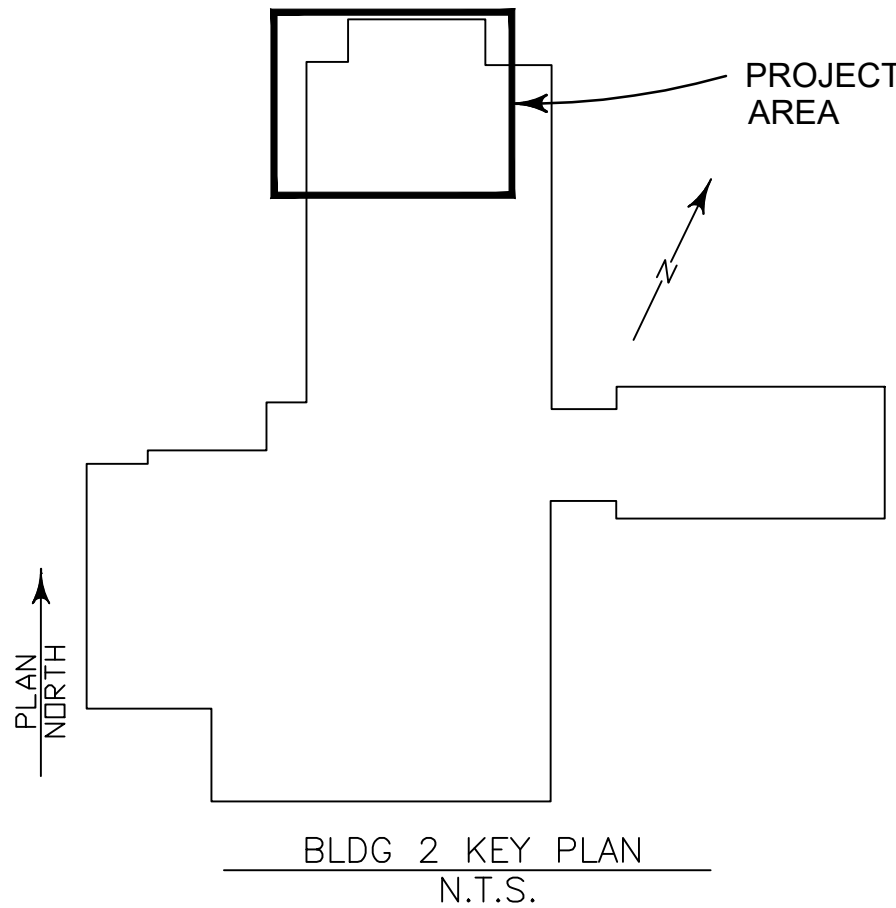
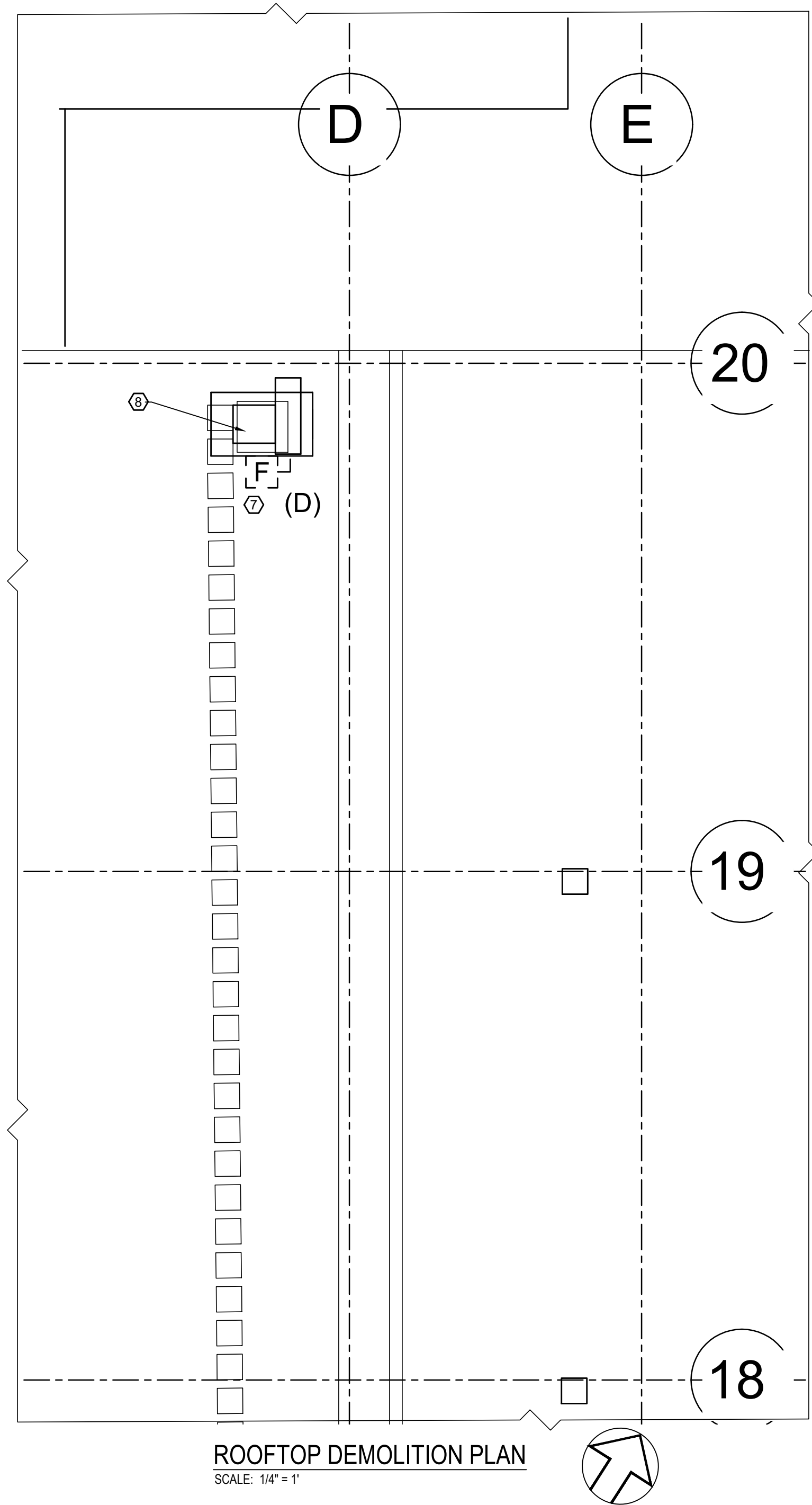
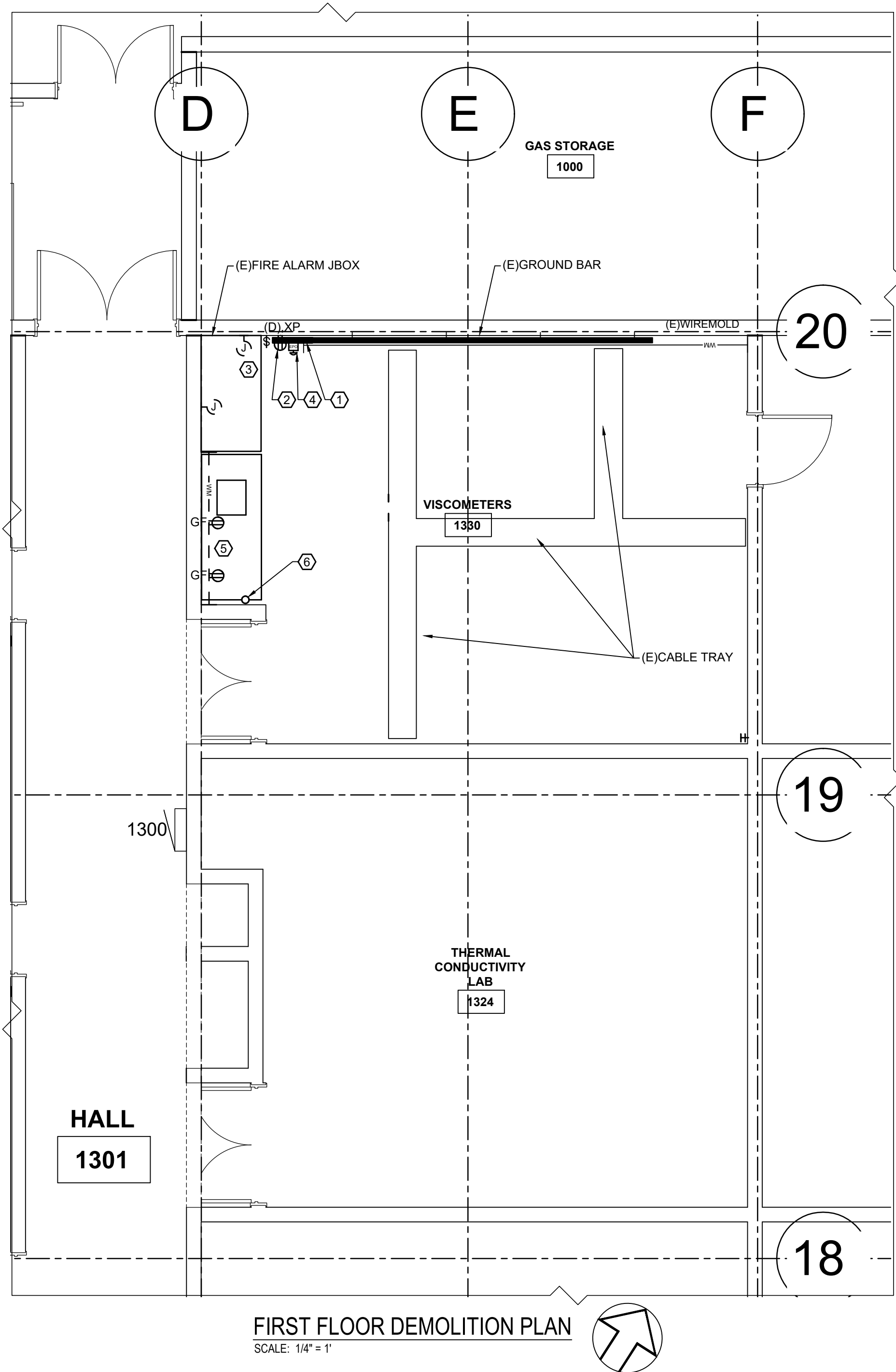
SHOULD ACTUAL CONDITIONS DEVIATE SUBSTANTIALLY FROM THOSE INDICATED ON THE DRAWING, CONTRACTOR SHALL NOTIFY ENGINEER AND REQUEST INSTRUCTIONS.

7.

NIST SHALL HAVE THE RIGHT TO SALVAGE ANY AND ALL MATERIALS AND EQUIPMENT OR PORTION THEREOF. ALL REMOVED EQUIPMENT MATERIALS NOT RETAINED BY NIST SHALL BE PROMPTLY REMOVED FROM NIST FACILITIES AND LEGALLY DISPOSED OF.

					
					
					
					
REVISION	DESCRIPTION	BY	SUPV.	DATE	

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY					
ENGINEERING, MAINTENANCE and SUPPORT SERVICES					
325 BROADWAY BOULDER, COLORADO 80305-3328					
BUILDING 2					
Room 1330 Walk-In Fume Hood					
ELECTRICAL					
DESIGN	DRAFTING	APPROVAL	DATE	SHEET NUMBER	
SJM	SJM		01/30/2026	E-1	
PROJECT NAME				of	
B2 - Room 1330 Walk-In Fume Hood					



GENERAL NOTES :

1. PLAN SHOWN IS DIAGRAMMATIC IN NATURE AND IS INTENDED FOR ILLUSTRATING SCOPE OF WORK. CONTRACTOR TO FIELD COORDINATE EXISTING
2. ROOM CONTAINS MANY LEGACY EXPLOSION PROOF DEVICES FROM PREVIOUS USERS. NEW DEVICES IN THIS ROOM WILL NOT BE REQUIRED TO HAVE EXPLOSION PROOF RATINGS.

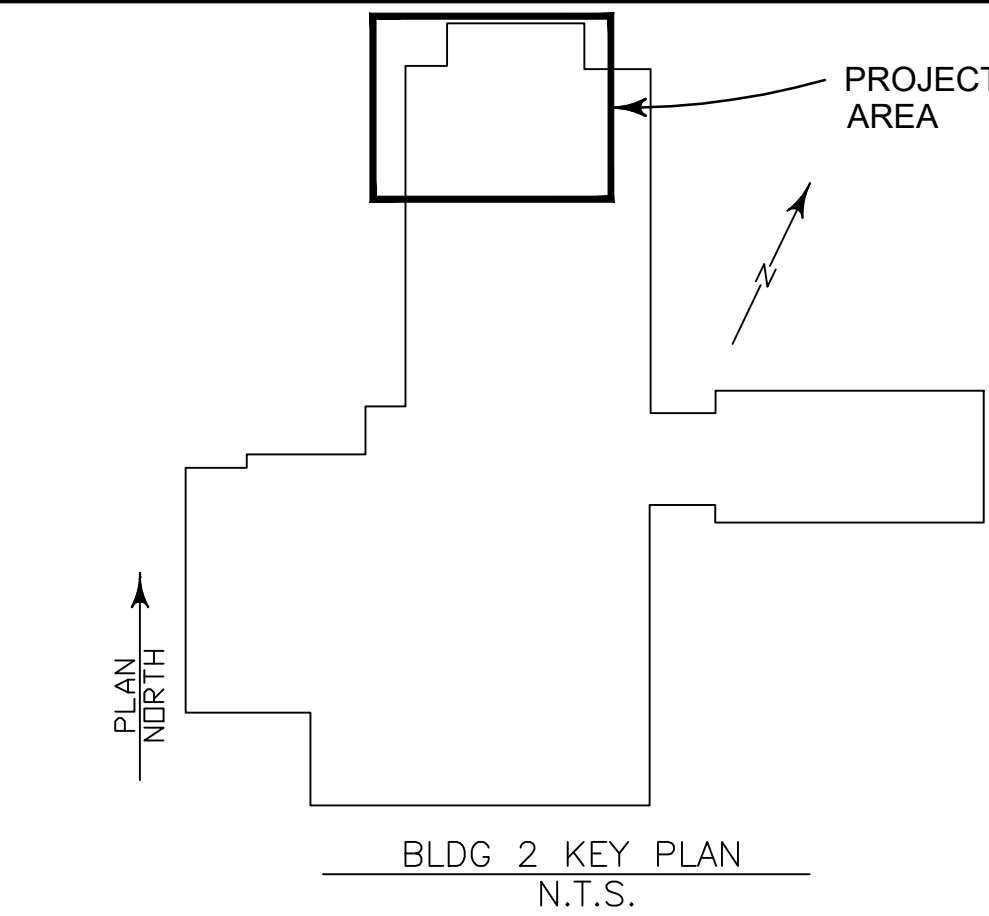
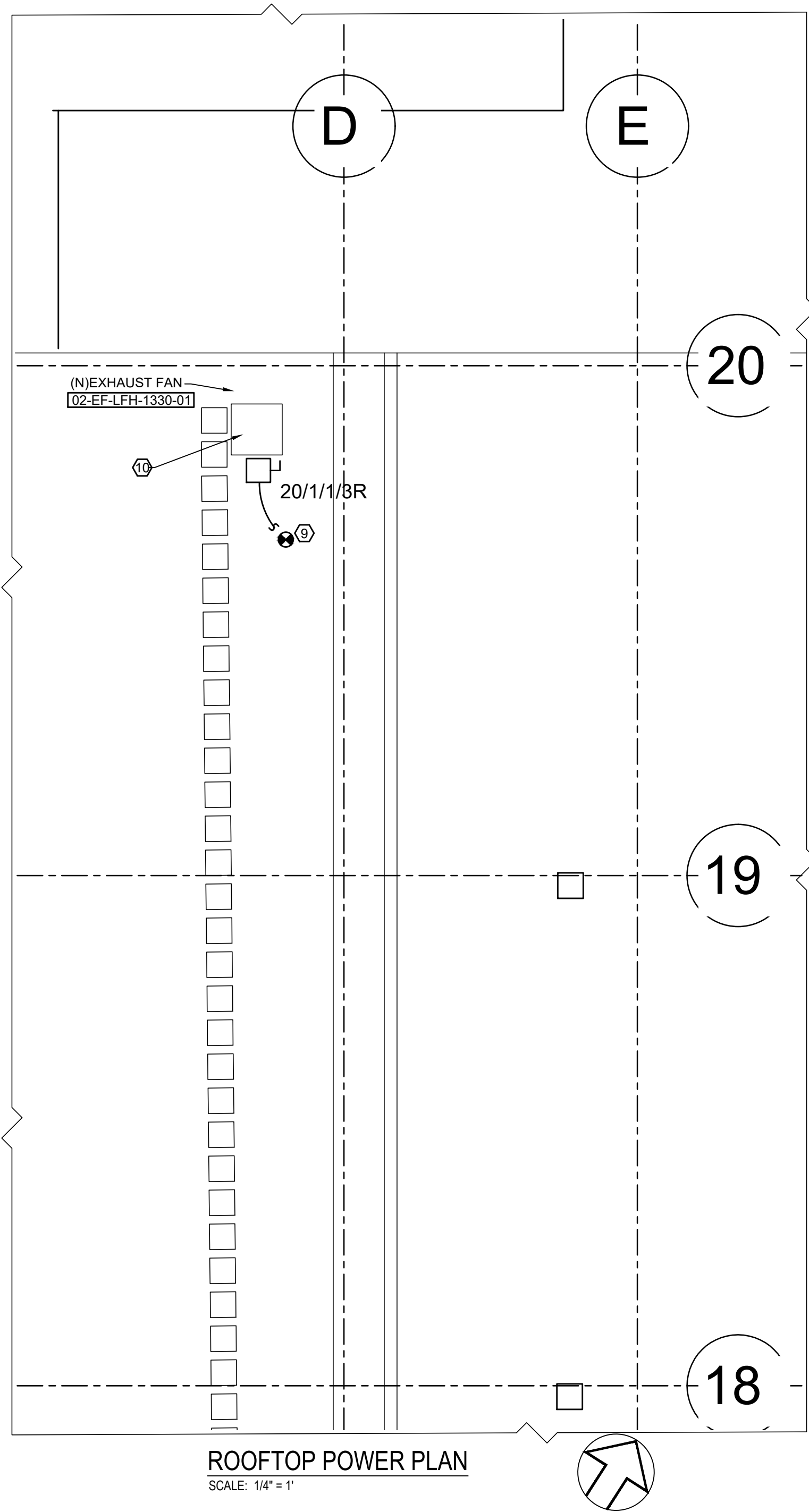
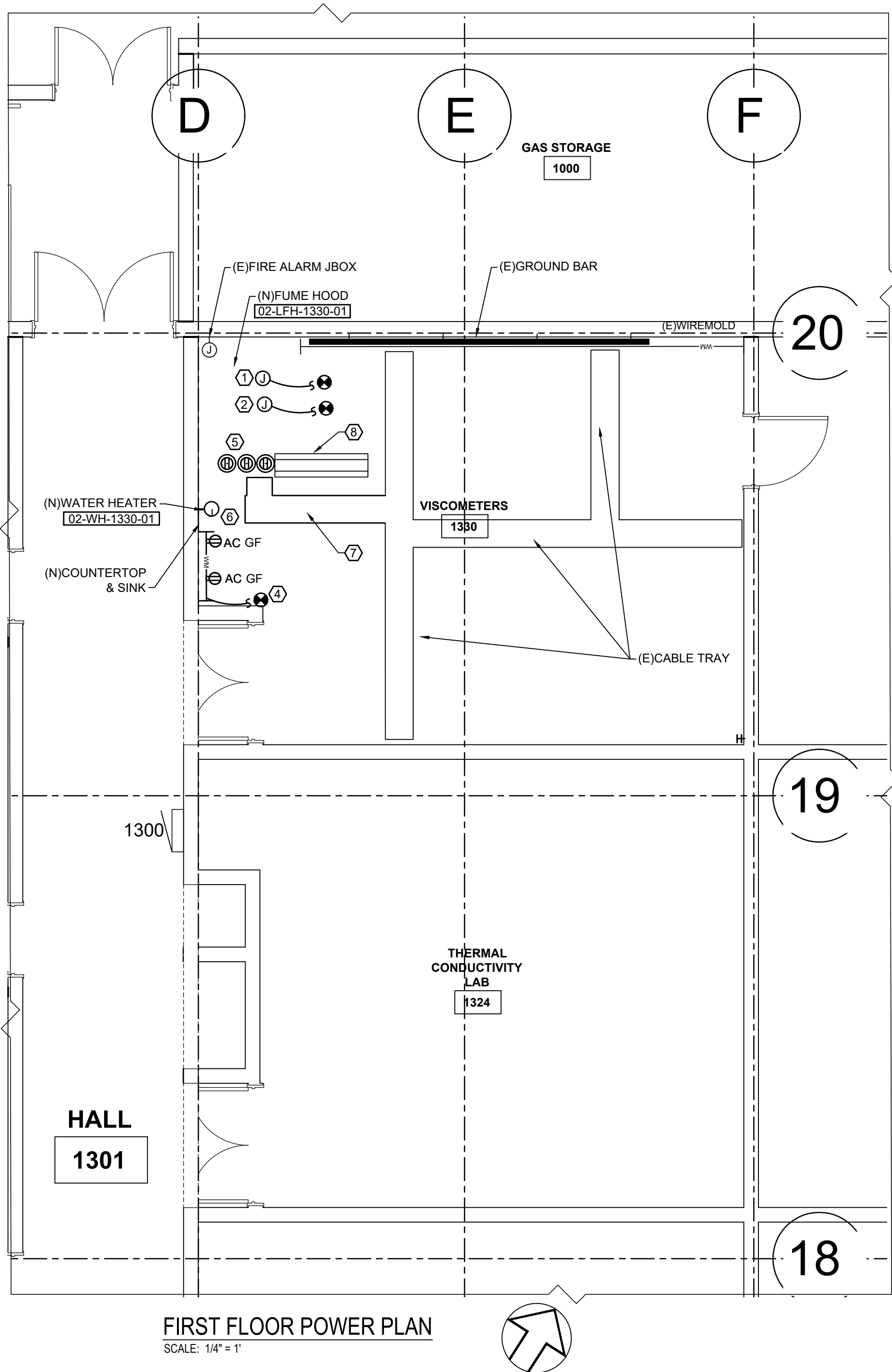
PLAN NOTES:

1. CONTRACTOR TO SHORTEN EXISTING GROUND BAR TO MAKE ROOM FOR NEW FUME HOOD. FIELD COORDINATE THE LENGTH OF GROUND BAR TO BE REMOVED. PROVIDE ADDITIONAL STANDOFF/WALL SUPPORTS AS NEEDED.
2. CONTRACTOR TO DEMO EXPLOSION PROOF RECEPTACLE AND ASSOCIATED CONDUIT AND CONDUCTORS BACK TO THE NEAREST DEVICE.
3. CONTRACTOR TO DEMO POWER CONNECTIONS TO EXISTING FUME HOOD AND EXISTING EXPLOSION PROOF SWITCH. REWORK EXISTING CONDUIT AND JUNCTION BOXES AS NEEDED TO REUSE EXISTING HOOD CIRCUIT(S) FOR NEW FUME HOOD.
4. CONTRACTOR TO DEMO EXISTING FUME HOOD PANIC BUTTON, CONDUIT AND CONDUCTORS BACK TO NEAREST JUNCTION BOX.
5. CONTRACTOR TO REMOVE EXISTING WIREMOLD ABOVE COUNTER. WIREMOLD AND EXISTING CIRCUITS TO BE SHORTEN AND REUSED ABOVE NEW COUNTERTOP.
6. CONTRACTOR TO REMOVE CONDUCTORS FROM EXISTING SEVERED CONDUIT BACK TO SOURCE AND CUT CONDUIT BACK FLUSH WITH THE FLOOR AND FILL.
7. CONTRACTOR TO DISCONNECT ELECTRICAL CONNECTIONS FROM EXHAUST FAN. CIRCUIT TO BE REUSED TO CONNECT TO NEW EXHAUST FAN. REWORK EXISTING CONDUITS AND WIRING FOR CONNECTION TO NEW EXHAUST FAN.
8. DISCONNECT EXISTING LIGHTNING PROTECTION SYSTEM FRO EXHAUST FAN/EXHAUST STACK AND PREP FOR RECONNECTION

SCALE: 1/8"=1'-0" (IN FEET)
0 4' 8' 16'

⚠				
⚠				
⚠				
⚠				
REVISION	DESCRIPTION	BY	SUPV.	DATE

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY ENGINEERING, MAINTENANCE and SUPPORT SERVICES 325 BROADWAY BOULDER, COLORADO 80305-3328				
BUILDING 2 Room 1330 Walk-In Fume Hood				
ELECTRICAL				
DESIGN SJM	DRAFTING SJM	APPROVAL	DATE 01/30/2026	SHEET NUMBER E-2 of
PROJECT NAME B2 - Room 1330 Walk-In Fume Hood				



- GENERAL NOTES :
- PLAN SHOWN IS DIAGRAMMATIC IN NATURE AND IS INTENDED FOR ILLUSTRATING SCOPE OF WORK. CONTRACTOR TO FIELD COORDINATE EXISTING
 - ROOM CONTAINS MANY LEGACY EXPLOSION PROOF DEVICES FROM PREVIOUS USERS. NEW DEVICES IN THIS ROOM WILL NOT BE REQUIRED TO HAVE EXPLOSION PROOF RATINGS.

- PLAN NOTES:
- CONTRACTOR TO REWORK EXISTING CONDUIT AND JUNCTION BOXES AS NEEDED TO REUSE EXISTING HOOD CIRCUIT(S) FOR NEW FUME HOOD INTERIOR LIGHTING AND CONTROLS.
 - CONTRACTOR TO REUSE EXISTING CONDUITS/PATHWAYS AND CONNECT NEW FUME HOOD TO BUILDING FIRE ALARM SYSTEM. FINAL CONNECTIONS BY QUALIFIED FIRE ALARM CONTRACTOR.
 - CONTRACTOR TO BOND NEW FUME HOOD, EXHAUST HOOD AND PROVIDE STATIC DISSIPATIVE CONNECTION TO FUME HOOD DUCTWORK. BONDING JUMPER TO BE CONNECTED TO EXISTING GROUNDING BAR IN THE ROOM. BONDING CONDUCTOR TO BE #6 AWG MINIMUM.
 - CONTRACTOR TO SHORTEN REMOVED WIREMOLD AND REINSTALL ABOVE COUNTER. CONTRACTOR TO PROVIDE NEW GFCI RECEPTACLES AND REWORK EXISTING WIRING AND CONDUITS TO RECONNECT EXISTING CIRCUITS TO WIREMOLD.
 - CONTRACTOR TO PROVIDE AND INSTALL THREE NEW RECEPTACLES WITH DEDICATED 120V, 20A CIRCUITS FROM PANEL 1300. COORDINATE EXACT LOCATION WITH FUME HOOD PASS-THRU. PROVIDE THREE NEW 20A 1-POLE BREKERS IN PANEL 1300. RECEPTACLES ARE TO BE UTILIZED BY USER EQUIPMENT INSIDE THE HOOD.
 - CONNECTION TO NEW INSTANTANEOUS WATER HEATER: CONTRACTOR TO PROVIDE NEW 208V, 20A, 1-PH CONNECTION TO WATER HEATER FROM PANEL PP6. PROVIDE NEW 20A 2-POLE BREKERS IN PANEL PP6.
 - CONTRACTOR TO PROVIDE AND INSTALL NEW CABLE TRAY FROM EXISTING CABLE TRAY WITH WATER FALLS AND AN ELBOW SECTION TO FUME HOOD PASS-THRU. CABLE TRAY TO BE BONDED PER CODE.
 - EXISTING LIGHT FIXTURE LOCATION IS IN CONFLICT WITH THE UP TRAVEL OF THE NEW FUME HOOD DOOR. CONTRACTOR TO REWORK EXISTING CONDUITS AND WIRING AS NEED TO RELOCATE EXISTING LIGHT FIXTURE OUT OF THE NEW FUM HOOD UPSWING.
 - EXHAUST FAN: CONTRACTOR TO PROVIDE NEW DISCONNECT, REWORK EXISTING CONDUITS/WIRING AND REUSE EXISTING CIRCUIT TO CONNECT NEW EXHAUST FAN.
 - REWORK AND CONNECT EXISTING LIGHTNING PROTECTION SYSTEM TO NEW EXHAUST FAN/EXHAUST STACK. WORK TO BE PERFORMED BY LICENSED LIGHTNING PROTECTION CONTRACTOR. CONTRACTOR TO RE-CERTIFY AND PROVIDE AS-BUILT OF MODIFICATIONS TO LIGHTNING PROTECTION SYSTEM.

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY ENGINEERING, MAINTENANCE and SUPPORT SERVICES 325 BROADWAY BOULDER, COLORADO 80305-3328				
BUILDING 2 Room 1330 Walk-In Fume Hood				
ELECTRICAL				
DESIGN SJM	DRAFTING SJM	APPROVAL	DATE 01/30/2026	SHEET NUMBER E-3
PROJECT NAME B2 - Room 1330 Walk-In Fume Hood				of

REVISION	DESCRIPTION	BY	SUPV.	DATE
1				
2				
3				
4				

SCALE: 1/8"=1'-0" (IN FEET)
0 4' 8' 16'